EXP 801 DS Wheat Heat Drought

— Ed Chaplin (USyd), *Apr–June 2025* —

Analysis of anatomical traits

Load packages

```
library(knitr)
#knitr::spin("Exp801.anat.anout.v1.r")
library(ggplot2)
library(asreml)
library(asremlPlus)
```

Set up constants and files

```
options(width=95)
load("../data/anat.Analyses.v1.RData")
source("Exp801.anat.anal.v1.stub.r")
```

Show the results

```
for (resp in responses.anat[kvec]) {
  showIndvModelResults(models[[resp]], response=resp)
}
```

```
#### Results for StomatalCount


  Final model converged



#### Formulae from asreml object

fixed: StomatalCount ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine +
     HWatering * HLine)
random: NULL
residual: ~ (dsum(~ units | Rm_Temperature))




####  Summary of the fitted variance parameters for StomatalCount

                           component std.error  z.ratio bound %ch
Rm_Temperature_7,Heat!R     10.44401  1.959970 5.328658     P   0
Rm_Temperature_8,Ambient!R  22.31792  4.114156 5.424665     P   0


####  Pseudo-anova table for fixed terms
```

Wald tests for fixed effects.
Response: StomatalCount

	Df	denDF	F.inc	F.con	Margin	Pr
(Intercept)	1	101.6	7756.0	7756.0		0.0000
ASide	1	101.6	2.2	2.0	B	0.1586
ASide:ABlock	2	101.7	5.7	6.0	C	0.0035
Rm_Temperature	1	104.0	0.1	0.1	A	0.7795
Rm_Temperature:AWatering	1	58.9	0.9	0.9	B	0.3359
Rm_Temperature:ALine	9	58.9	0.9	0.9	B	0.5552
Rm_Temperature:AWatering:ALine	9	58.9	0.8	0.8	C	0.5940
Rm_Temperature:HWatering	1	56.8	0.9	1.1	B	0.3067
Rm_Temperature:HLine	9	56.8	7.1	7.1	B	0.0000
Rm_Temperature:HWatering:HLine	9	56.8	1.7	1.7	C	0.1123

Sequence of model investigations for StomatalCount

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

	terms	DF	denDF	p	AIC	BIC	action
1	Initial model	43	1.0	NA	649.460213	784.491998	Starting model
2	Start with residual heterogeneity in Watering & Rm_Temperature for Residual	NA	NA	NA	NA	NA	Changed residual
3	Try homogeneity in Rm_Temperature for Residual	0	-2.0	NA	4.283150	-1.224031	Unswapped
4	Try homogeneity in Watering for Residual	0	-2.0	NA	-3.861354	-9.368534	Swapped
5	Room:Side	1	NA	NA	NA	NA	Boundary
6	Room:Side:Location	1	NA	NA	NA	NA	Boundary
7	Room:Side:Location:SubUnit	1	NA	NA	NA	NA	Boundary
8	Fix bound variance components	0	0.0	NA	0.000000	0.000000	Swapped
9	Rm_Temperature:HWatering:HLine	9	56.8	0.1123	NA	NA	Nonsignificant
10	Rm_Temperature:HLine	9	56.8	0.0000	NA	NA	Significant
11	Rm_Temperature:HWatering	1	56.8	0.3067	NA	NA	Nonsignificant
12	Rm_Temperature:AWatering:ALine	9	58.9	0.5940	NA	NA	Nonsignificant
13	Rm_Temperature:ALine	9	58.9	0.5552	NA	NA	Nonsignificant
14	Rm_Temperature:AWatering	1	58.9	0.3359	NA	NA	Nonsignificant

Information criteria for spatial models

```
NULL

Chosen terms
[1] "Rm_Temperature:HLine"

LSDs
          c   minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 4.526069 5.683925 6.616287    5.683925   0.2037072       12       22

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`


#### Results for StomatalDensity


  Final model converged



#### Formulae from asreml object

fixed: StomatalDensity ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine +
     HWatering * HLine)
random: NULL
residual: ~ (dsum(~ units | Rm_Temperature))




####  Summary of the fitted variance parameters for StomatalDensity

                           component std.error  z.ratio bound %ch
Rm_Temperature_7,Heat!R     39.48046  7.409081 5.328658     P   0
Rm_Temperature_8,Ambient!R  84.36620 15.552334 5.424665     P   0


####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: StomatalDensity

                                Df denDF  F.inc  F.con Margin     Pr
(Intercept)                      1 101.6 7756.0 7756.0        0.0000
ASide                            1 101.6    2.2    2.0      B 0.1586
ASide:ABlock                     2 101.7    5.7    6.0      C 0.0035
Rm_Temperature                   1 104.0    0.1    0.1      A 0.7795
Rm_Temperature:AWatering         1  58.9    0.9    0.9      B 0.3359
Rm_Temperature:ALine             9  58.9    0.9    0.9      B 0.5552
Rm_Temperature:AWatering:ALine   9  58.9    0.8    0.8      C 0.5940
Rm_Temperature:HWatering         1  56.8    0.9    1.1      B 0.3067
Rm_Temperature:HLine             9  56.8    7.1    7.1      B 0.0000
Rm_Temperature:HWatering:HLine   9  56.8    1.7    1.7      C 0.1123
```

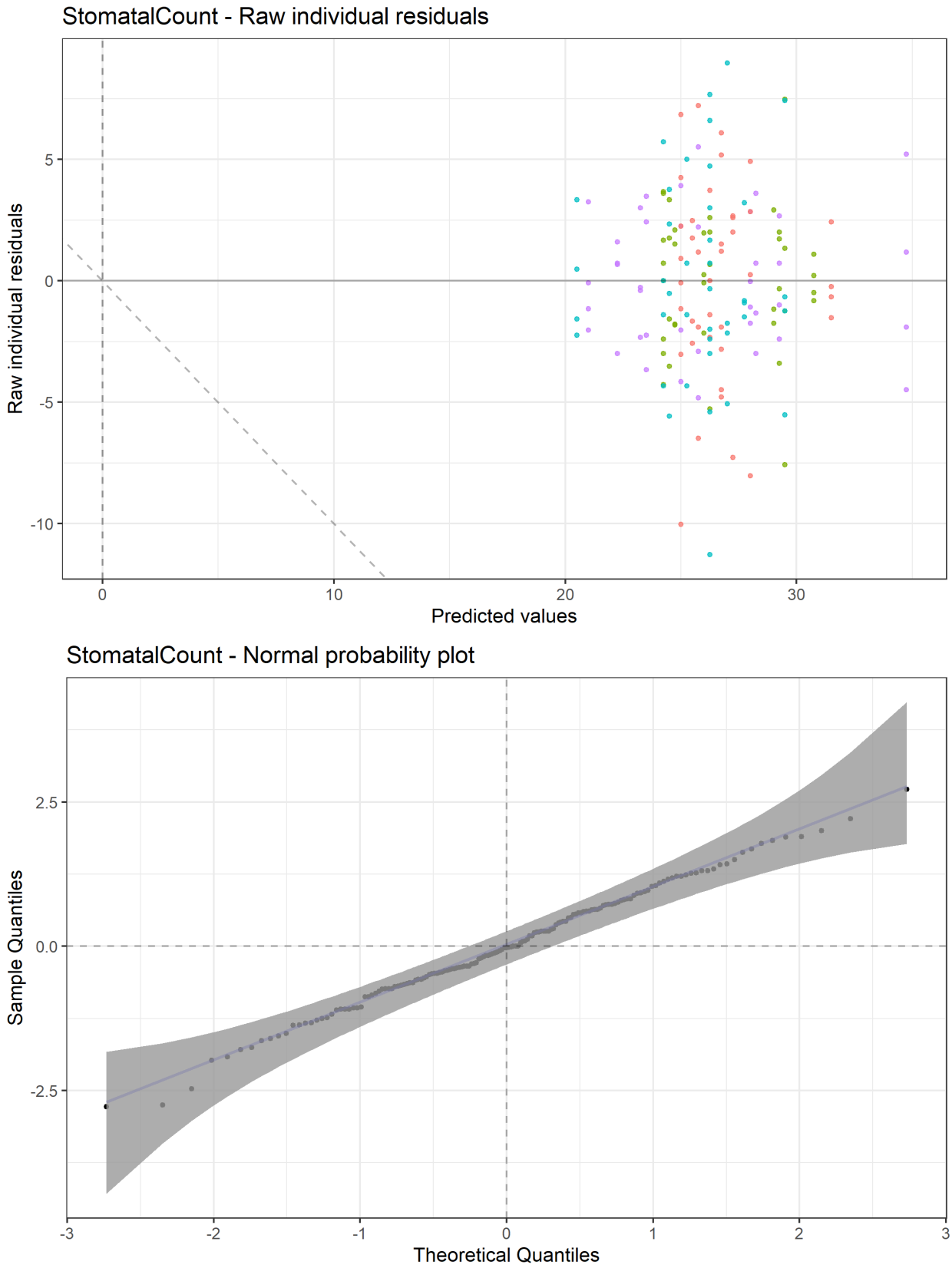


Figure 1: plot of chunk show-v1

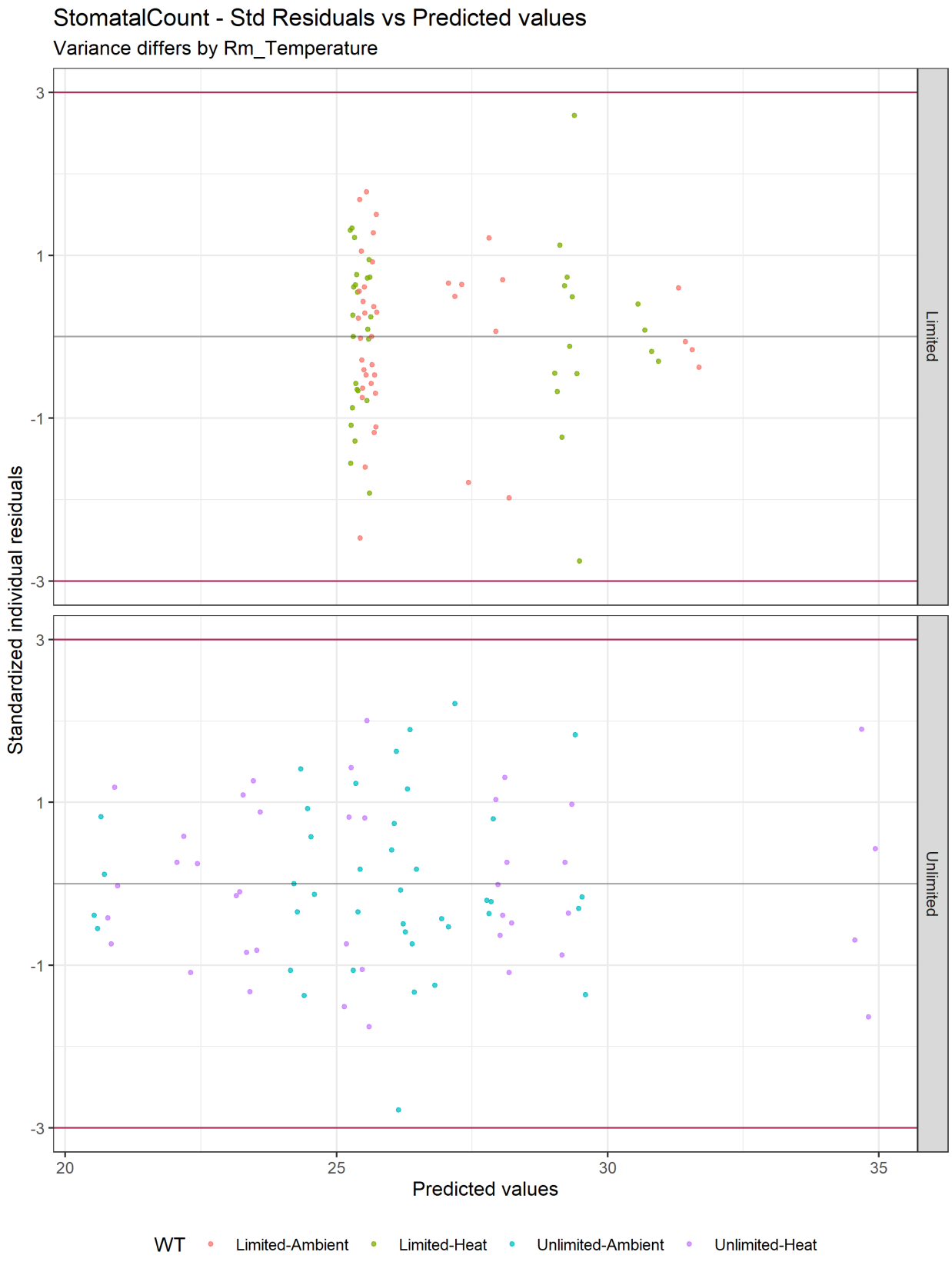


Figure 2: plot of chunk show-v1

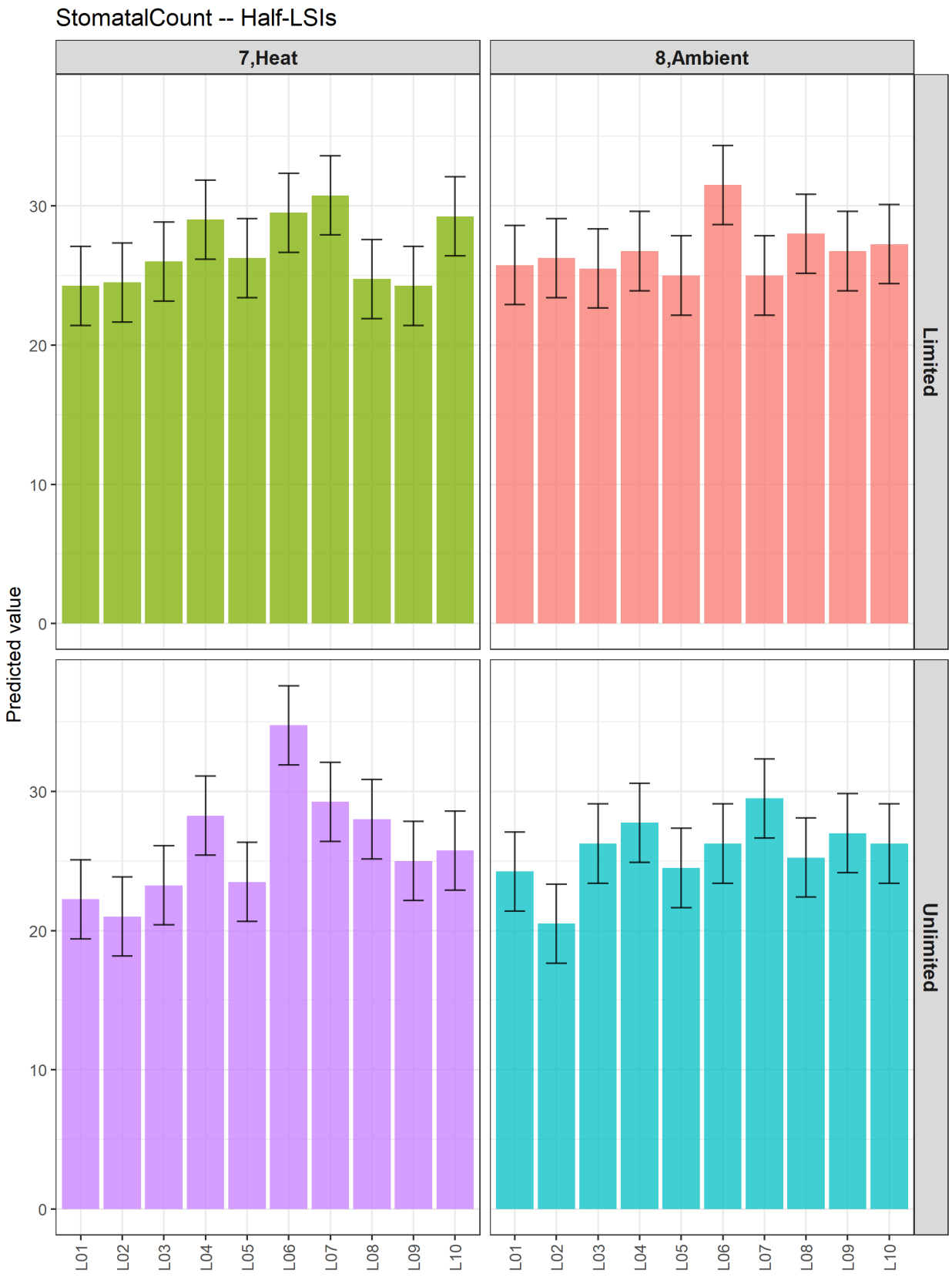


Figure 3: plot of chunk show-v1

```
####  Sequence of model investigations for StomatalDensity

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

                                                                                terms DF denDF      p
1                                                                       Initial model 43   1.0     NA
2  Start with residual heterogeneity in Watering & Rm_Temperature for Residual NA    NA     NA
3                                 Try homogeneity in Rm_Temperature for Residual  0  -2.0     NA
4                                       Try homogeneity in Watering for Residual  0  -2.0     NA
5                                                                    Room:Side  1    NA     NA
6                                                            Room:Side:Location  1    NA     NA
7                                                       Room:Side:Location:SubUnit  1    NA     NA
8                                                    Fix bound variance components  0   0.0     NA
9                                                   Rm_Temperature:HWatering:HLine  9  56.8 0.1123
10                                                            Rm_Temperature:HLine  9  56.8 0.0000
11                                                          Rm_Temperature:HWatering  1  56.8 0.3067
12                                                  Rm_Temperature:AWatering:ALine  9  58.9 0.5940
13                                                            Rm_Temperature:ALine  9  58.9 0.5552
14                                                          Rm_Temperature:AWatering  1  58.9 0.3359
          AIC        BIC           action
1  860.894764 995.926549   Starting model
2          NA         NA Changed residual
3    4.283150  -1.224031        Unswapped
4   -3.861354  -9.368534          Swapped
5          NA         NA         Boundary
6          NA         NA         Boundary
7          NA         NA         Boundary
8    0.000000   0.000000          Swapped
9          NA         NA   Nonsignificant
10         NA         NA      Significant
11         NA         NA   Nonsignificant
12         NA         NA   Nonsignificant
13         NA         NA   Nonsignificant
14         NA         NA   Nonsignificant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine"

LSDs
          c   minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 8.799917 11.05111 12.86387    11.05111   0.2037072       12       22

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`


#### Results for StomatalArea
```

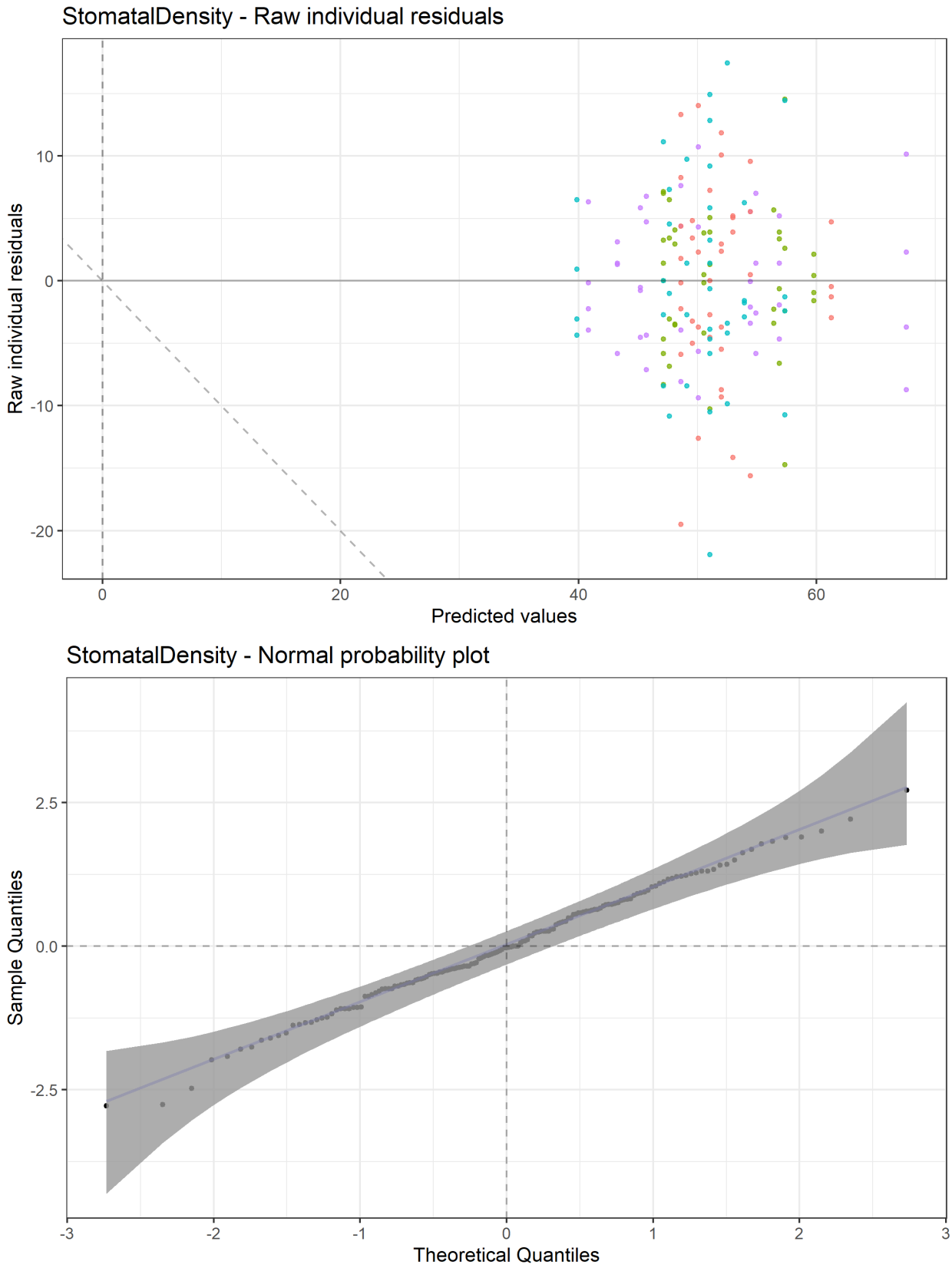


Figure 4: plot of chunk show-v1

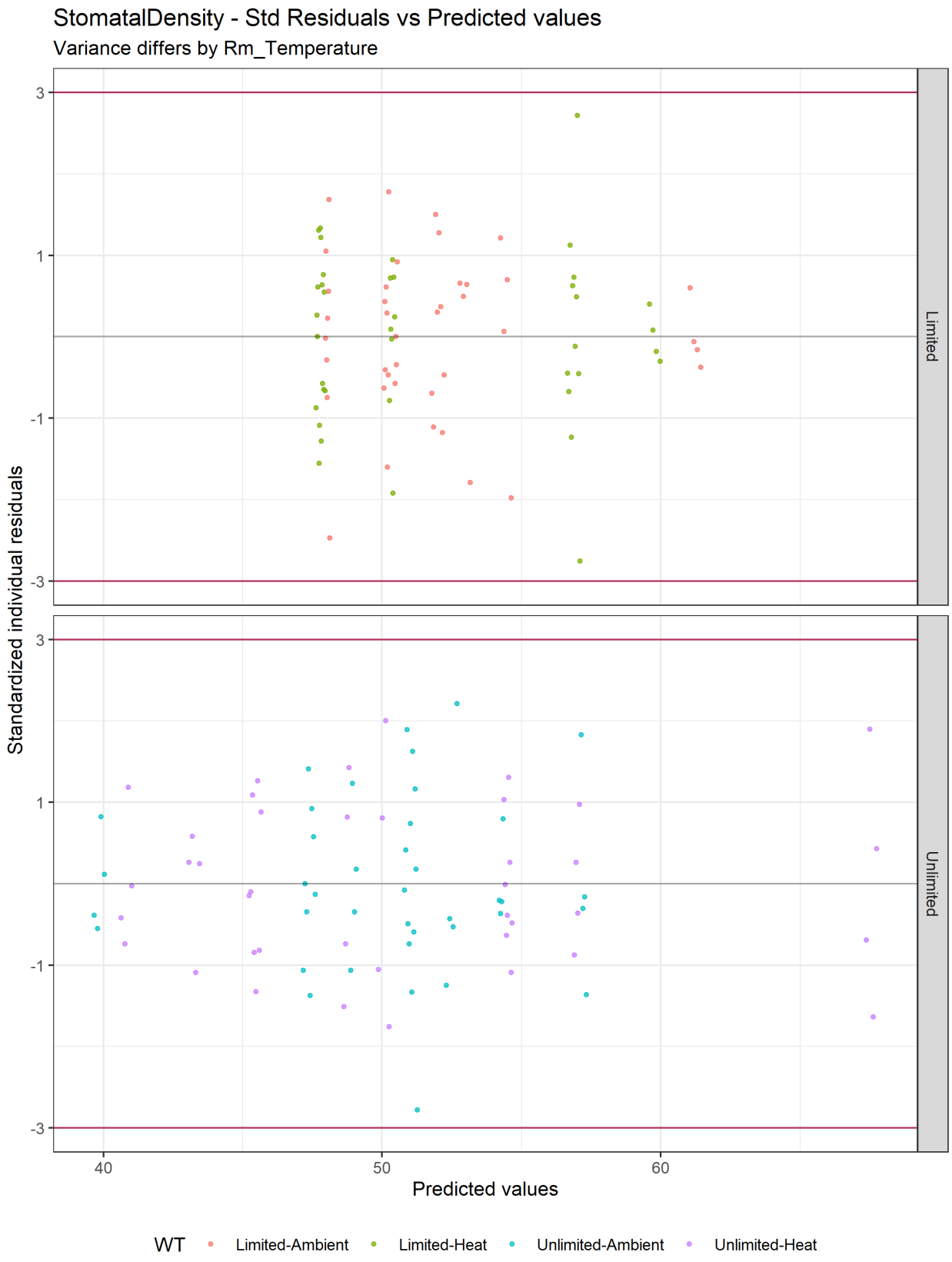


Figure 5: plot of chunk show-v1

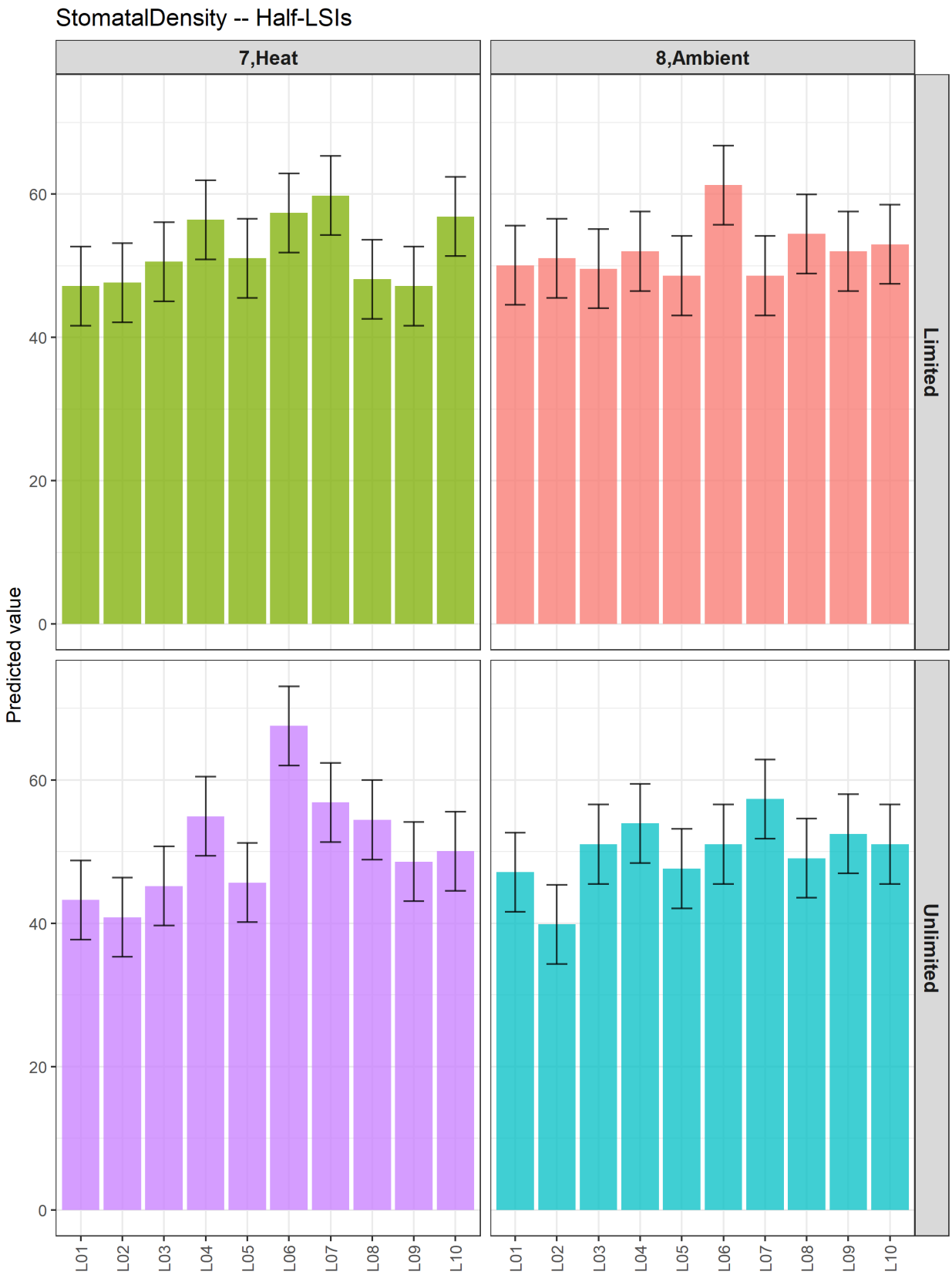


Figure 6: plot of chunk show-v1

```
  Final model converged


#### Formulae from asreml object

fixed: StomatalArea ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine +
     HWatering * HLine)
random: ~ Room:Side + Room:Side:Location:SubUnit +
     Room:Side:Location:SubUnit:Cell
residual: ~ (dsum(~ units | Rm_Temperature))




####  Summary of the fitted variance parameters for StomatalArea

                                  component std.error    z.ratio bound %ch
Room:Side                          73.50967  2849.146  0.0258006     P   0
Room:Side:Location:SubUnit       4251.30086  3683.569  1.1541254     P   0
Room:Side:Location:SubUnit:Cell 27684.33016  4167.307  6.6432184     P   0
Rm_Temperature_7,Heat!R         80242.34575  2530.630 31.7084472     P   0
Rm_Temperature_8,Ambient!R      63677.22734  2003.859 31.7772984     P   0


####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: StomatalArea

                              Df denDF  F.inc  F.con Margin     Pr
(Intercept)                    1   1.0 5643.0 5643.0        0.0085
ASide                          1   1.0    2.7    2.6      B 0.3560
ASide:ABlock                   2   8.0    0.1    0.1      C 0.9477
Rm_Temperature                 1   1.0    0.4    0.4      A 0.6300
Rm_Temperature:AWatering       1   7.9    2.8    2.8      B 0.1332
Rm_Temperature:ALine           9 104.9    3.4    3.4      B 0.0010
Rm_Temperature:AWatering:ALine 9 104.9    1.6    1.6      C 0.1399
Rm_Temperature:HWatering       1   8.1    1.7    1.5      B 0.2523
Rm_Temperature:HLine           9 109.2    3.3    3.3      B 0.0013
Rm_Temperature:HWatering:HLine 9 109.3    0.3    0.3      C 0.9723


####  Sequence of model investigations for StomatalArea

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

                                                                    terms DF denDF      p
1                                                             Initial model 43   4.0     NA
2  Start with residual heterogeneity in Watering & Rm_Temperature for Residual NA    NA     NA
3                              Try homogeneity in Rm_Temperature for Residual  0  -2.0     NA
4                                    Try homogeneity in Watering for Residual  0  -2.0     NA
```

```
5                                                          Room:Side:Location  1    NA     NA
6                                                   Fix bound variance components  0   0.0     NA
7                                                 Rm_Temperature:HWatering:HLine  9 109.3 0.9723
8                                                           Rm_Temperature:HLine  9 109.2 0.0013
9                                                 Rm_Temperature:HWatering:Line  1   8.1 0.2523
10                                              Rm_Temperature:AWatering:ALine  9 104.9 0.1399
11                                                          Rm_Temperature:ALine  9 104.9 0.0010
12                                                Rm_Temperature:AWatering:Line  1   7.9 0.1332
            AIC         BIC           action
1  51482.844762 51780.83498   Starting model
2            NA          NA Changed residual
3     23.589533    10.92973        Unswapped
4     -3.121447   -15.78125          Swapped
5            NA          NA         Boundary
6      0.000000     0.00000          Swapped
7            NA          NA   Nonsignificant
8            NA          NA      Significant
9            NA          NA   Nonsignificant
10           NA          NA   Nonsignificant
11           NA          NA      Significant
12           NA          NA   Nonsignificant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine" "Rm_Temperature:ALine"

LSDs
          c   minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 239.5736 256.0375 279.8144    256.0375  0.09286501        5        1

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`


#### Results for StomatalWidth


  Final model converged



#### Formulae from asreml object

fixed: StomatalWidth ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine +
    HWatering * HLine)
random: ~ Room:Side + Room:Side:Location:SubUnit +
    Room:Side:Location:SubUnit:Cell
residual: ~ dsum(~ units | WateringRm_Temperature)
```

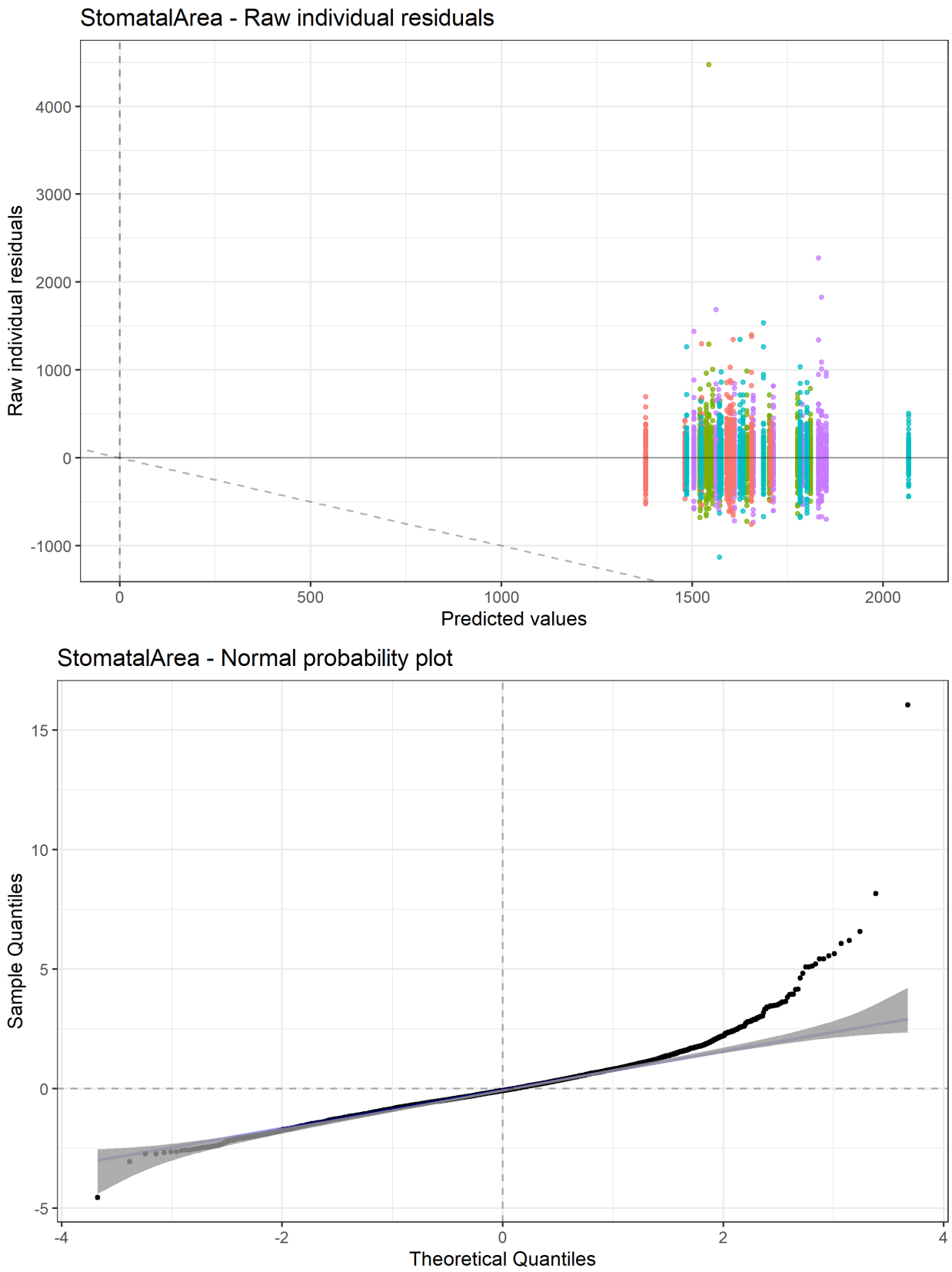


Figure 7: plot of chunk show-v1

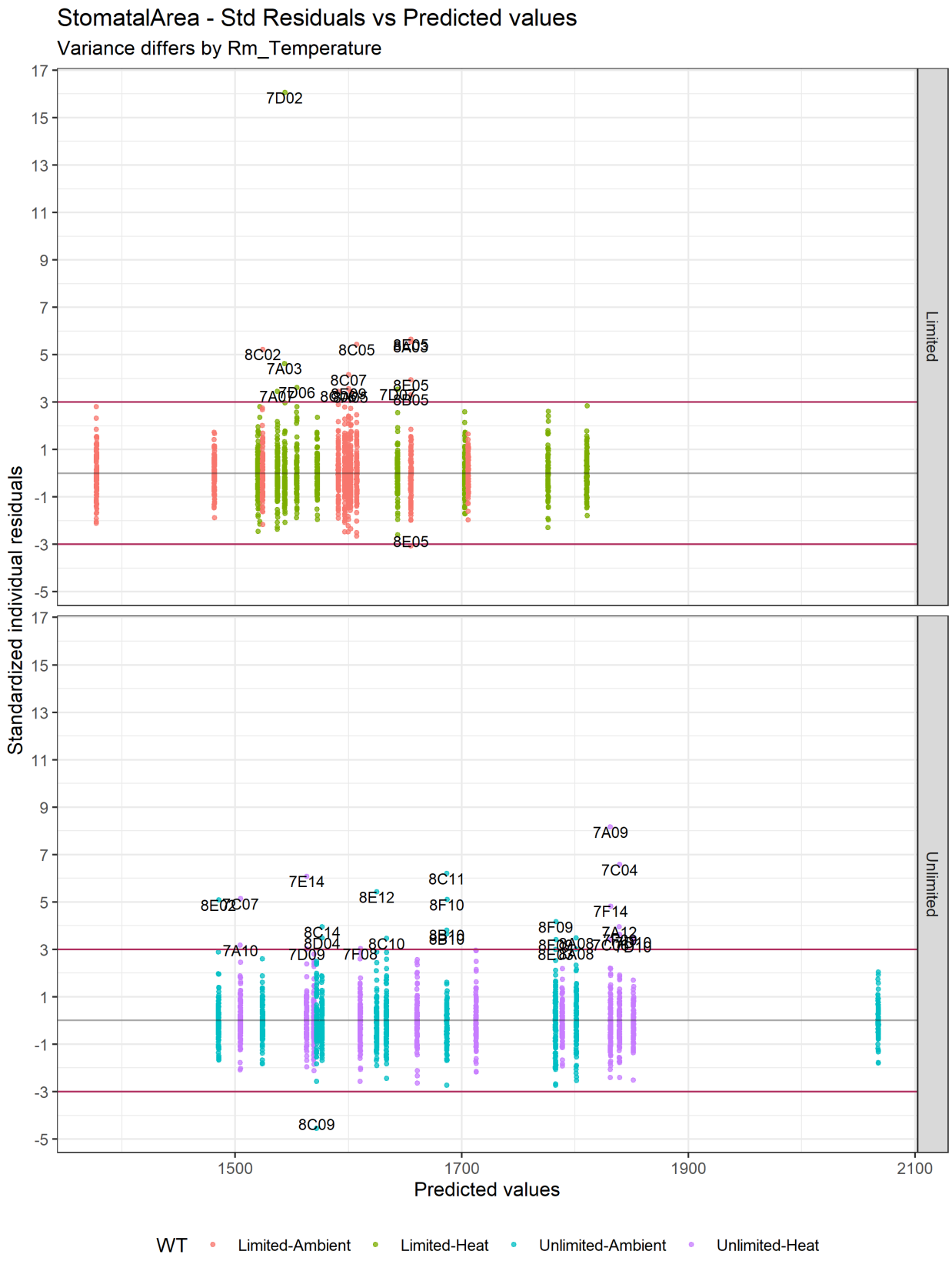


Figure 8: plot of chunk show-v1

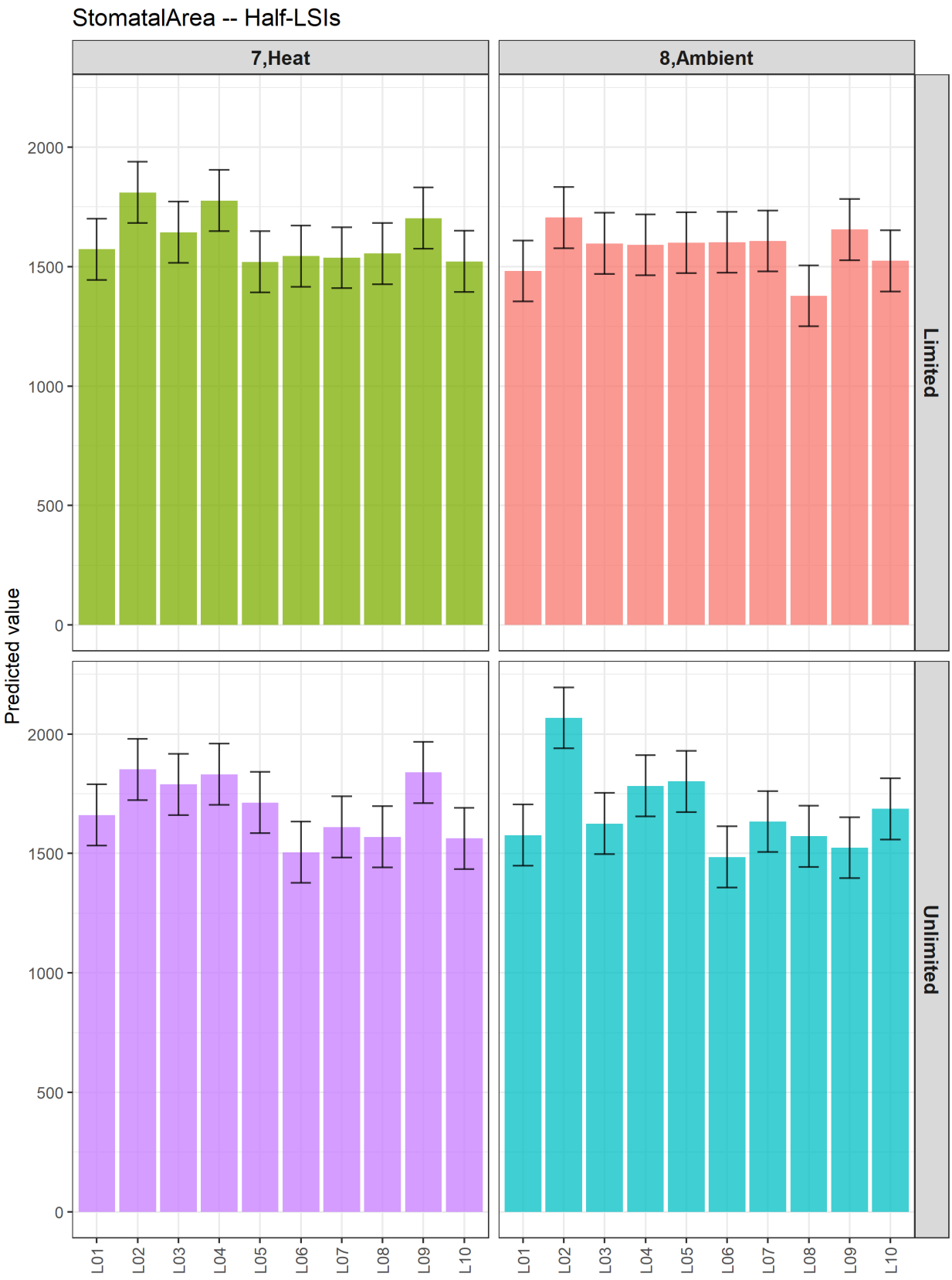


Figure 9: plot of chunk show-v1

```
####  Summary of the fitted variance parameters for StomatalWidth

                                          component std.error    z.ratio bound %ch
Room:Side                                 0.1484938 0.5902575  0.2515746     P   0
Room:Side:Location:SubUnit                0.1077919 0.5306323  0.2031386     P   0
Room:Side:Location:SubUnit:Cell           8.4966975 1.2553291  6.7685019     P   0
WateringRm_Temperature_Limited7,Heat!R   17.4083994 0.7763376 22.4237499     P   0
WateringRm_Temperature_Limited8,Ambient!R 15.6030113 0.6869376 22.7138713     P   0
WateringRm_Temperature_Unlimited7,Heat!R  19.0023201 0.8478592 22.4121182     P   0
WateringRm_Temperature_Unlimited8,Ambient!R 17.2784271 0.7774182 22.2253946     P   0


####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: StomatalWidth

                              Df denDF   F.inc   F.con Margin     Pr
(Intercept)                    1   1.0 11500.0 11500.0        0.0059
ASide                          1   1.0     2.4     2.2      B 0.3760
ASide:ABlock                   2   8.0     0.1     0.1      C 0.9133
Rm_Temperature                 1   1.0     0.3     0.3      A 0.6759
Rm_Temperature:AWatering       1   7.9     8.4     8.4      B 0.0200
Rm_Temperature:ALine           9 106.1     2.4     2.4      B 0.0160
Rm_Temperature:AWatering:ALine 9 106.1     2.3     2.3      C 0.0207
Rm_Temperature:HWatering       1   8.1     7.5     6.7      B 0.0322
Rm_Temperature:HLine           9 107.6     3.2     3.2      B 0.0019
Rm_Temperature:HWatering:HLine 9 107.7     0.3     0.3      C 0.9776


####  Sequence of model investigations for StomatalWidth

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

                                                                    terms DF denDF      p
1                                                                Initial model 43   4.0     NA
2  Start with residual heterogeneity in Watering & Rm_Temperature for Residual NA    NA     NA
3                                Try homogeneity in Rm_Temperature for Residual  0  -2.0     NA
4                                      Try homogeneity in Watering for Residual  0  -2.0     NA
5                                                  Room:Side:Location  1    NA     NA
6                                             Fix bound variance components  0   0.0     NA
7                                           Rm_Temperature:HWatering:HLine  9 107.7 0.9776
8                                                    Rm_Temperature:HLine  9 107.6 0.0019
9                                                Rm_Temperature:HWatering  1   8.1 0.0322
10                                          Rm_Temperature:AWatering:ALine  9 106.1 0.0207
            AIC         BIC           action
1  1.660484e+04 16902.82530   Starting model
2            NA          NA Changed residual
3  1.303480e+00   -11.35632        Unswapped
4  5.531296e-01   -12.10667        Unswapped
5            NA          NA         Boundary
6  0.000000e+00     0.00000          Swapped
```

```
7            NA          NA   Nonsignificant
8            NA          NA      Significant
9            NA          NA      Significant
10           NA          NA      Significant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine"           "Rm_Temperature:HWatering"
[3] "Rm_Temperature:AWatering:ALine"

LSDs
          c   minLSD meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 4.166099 4.27011 4.640635     4.27011  0.08677171        4        0

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`



#### Results for StomatalLength


  Final model converged



#### Formulae from asreml object

fixed: StomatalLength ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine +
     HWatering * HLine)
random: ~ Room:Side:Location + Room:Side:Location:SubUnit +
     Room:Side:Location:SubUnit:Cell
residual: ~ dsum(~ units | WateringRm_Temperature)




####  Summary of the fitted variance parameters for StomatalLength

                                           component std.error    z.ratio bound %ch
Room:Side:Location                          0.09595782  1.723767  0.0556675     P   0
Room:Side:Location:SubUnit                  1.39560309  1.936181  0.7208019     P   0
Room:Side:Location:SubUnit:Cell            17.66577639  2.620247  6.7420269     P   0
WateringRm_Temperature_Limited7,Heat!R     51.79731433  2.310253 22.4206245     P   0
WateringRm_Temperature_Limited8,Ambient!R  36.03829178  1.587711 22.6982704     P   0
WateringRm_Temperature_Unlimited7,Heat!R   33.61434186  1.499633 22.4150417     P   0
WateringRm_Temperature_Unlimited8,Ambient!R 33.48591790  1.505128 22.2478926     P   0


####  Pseudo-anova table for fixed terms
```

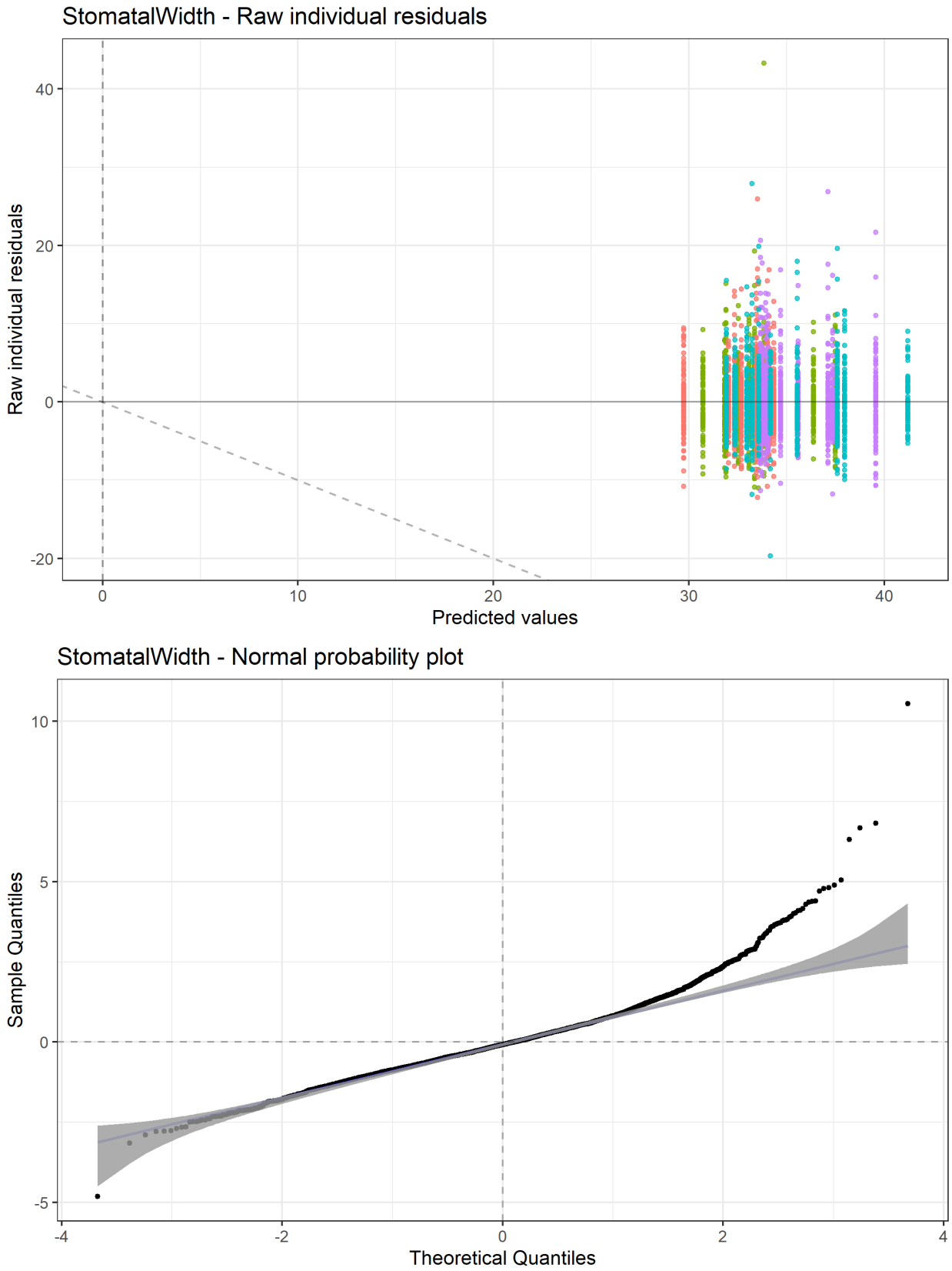


Figure 10: plot of chunk show-v1

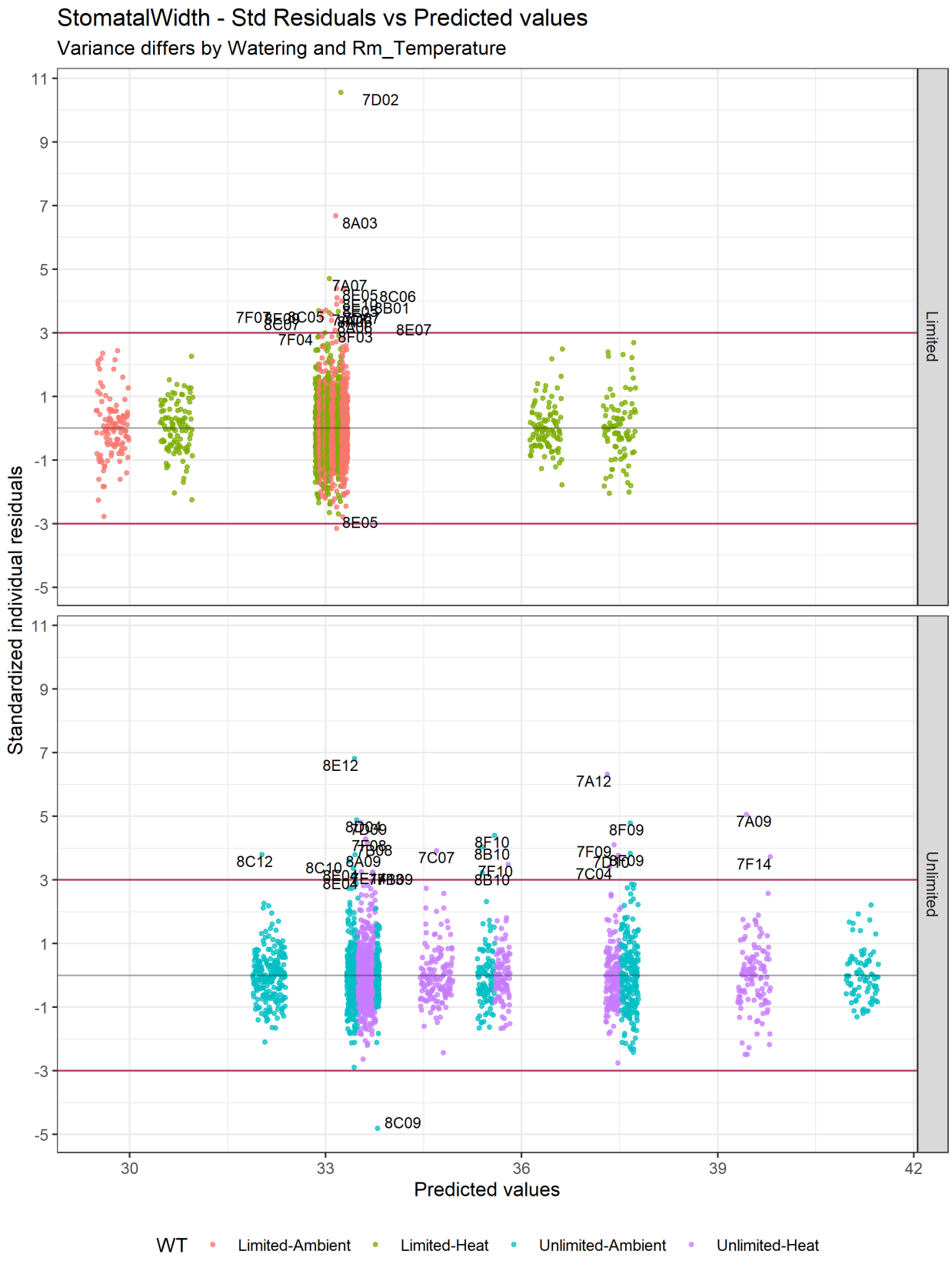


Figure 11: plot of chunk show-v1

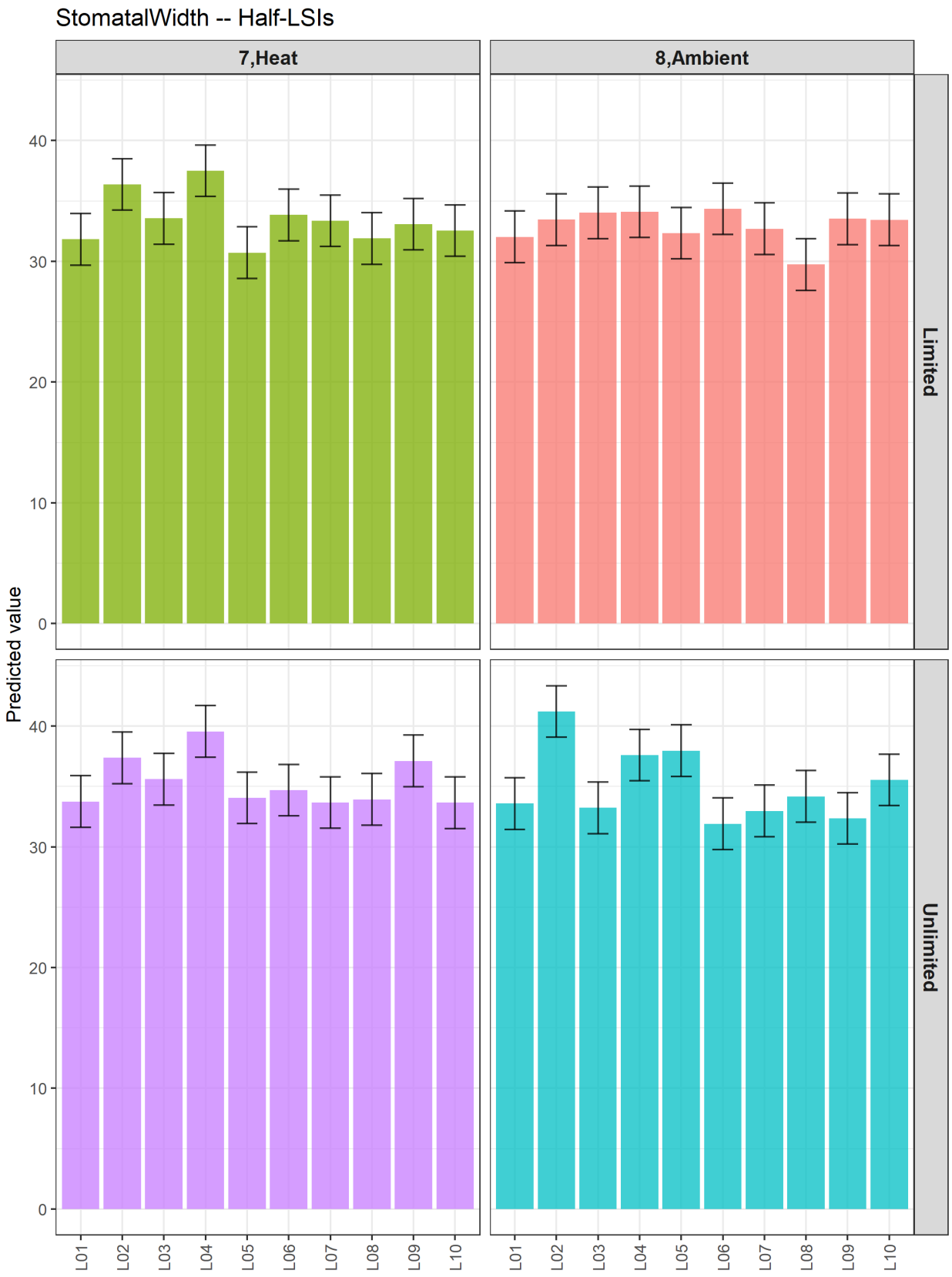


Figure 12: plot of chunk show-v1

Wald tests for fixed effects.
Response: StomatalLength

	Df	denDF	F.inc	F.con	Margin	Pr
(Intercept)	1	3.0	17670.0	17670.0		0.0000
ASide	1	3.0	0.3	0.3	B	0.6115
ASide:ABlock	2	3.0	0.6	0.6	C	0.6007
Rm_Temperature	1	3.0	0.3	0.3	A	0.6352
Rm_Temperature:AWatering	1	5.9	0.1	0.1	B	0.8285
Rm_Temperature:ALine	9	105.5	1.5	1.5	B	0.1748
Rm_Temperature:AWatering:ALine	9	105.5	0.5	0.5	C	0.8681
Rm_Temperature:HWatering	1	6.1	0.1	0.1	B	0.7591
Rm_Temperature:HLine	9	108.6	3.2	3.2	B	0.0019
Rm_Temperature:HWatering:HLine	9	108.8	0.5	0.5	C	0.8885

Sequence of model investigations for StomatalLength

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

	terms	DF	denDF	p	AIC	BIC	action
1	Initial model	43	4.0	NA	19973.47943	20271.46964	Starting model
2	Start with residual heterogeneity in Watering & Rm_Temperature for Residual	NA	NA	NA	NA	NA	Changed residual
3	Try homogeneity in Rm_Temperature for Residual	0	-2.0	NA	29.34297	16.68317	Unswapped
4	Try homogeneity in Watering for Residual	0	-2.0	NA	44.00225	31.34245	Unswapped
5	Room:Side	1	NA	NA	NA	NA	Boundary
6	Fix bound variance components	0	0.0	NA	0.00000	0.00000	Swapped
7	Rm_Temperature:HWatering:HLine	9	108.8	0.8885	NA	NA	Nonsignificant
8	Rm_Temperature:HLine	9	108.6	0.0019	NA	NA	Significant
9	Rm_Temperature:HWatering	1	6.1	0.7591	NA	NA	Nonsignificant
10	Rm_Temperature:AWatering:ALine	9	105.5	0.8681	NA	NA	Nonsignificant
11	Rm_Temperature:ALine	9	105.5	0.1748	NA	NA	Nonsignificant
12	Rm_Temperature:AWatering	1	5.9	0.8285	NA	NA	Nonsignificant

Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine"

```
LSDs
          c   minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 6.000523 6.271869 6.838396    6.271869  0.09032831        0        0

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`


#### Results for PoreArea


  Final model converged



#### Formulae from asreml object

fixed: PoreArea ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine + HWatering *
     HLine)
random: ~ Room:Side:Location + Room:Side:Location:SubUnit +
     Room:Side:Location:SubUnit:Cell
residual: ~ dsum(~ units | WateringRm_Temperature)




####  Summary of the fitted variance parameters for PoreArea

                                          component std.error    z.ratio bound %ch
Room:Side:Location                         143.8787  1391.887  0.1033695     P   0
Room:Side:Location:SubUnit                1063.3596  1513.633  0.7025212     P   0
Room:Side:Location:SubUnit:Cell          13603.3456  2084.202  6.5268835     P   0
WateringRm_Temperature_Limited7,Heat!R   91606.4979  4083.603 22.4327622     P   0
WateringRm_Temperature_Limited8,Ambient!R 32017.5582  1410.684 22.6964752     P   0
WateringRm_Temperature_Unlimited7,Heat!R  29987.3433  1337.760 22.4160790     P   0
WateringRm_Temperature_Unlimited8,Ambient!R 28782.4611  1293.710 22.2479966     P   0


####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: PoreArea

                            Df denDF  F.inc  F.con Margin     Pr
(Intercept)                  1   3.0 4601.0 4601.0        0.0000
ASide                        1   3.0    0.3    0.3      B 0.6178
ASide:ABlock                 2   3.0    0.6    0.6      C 0.6088
Rm_Temperature               1   3.0    0.3    0.3      A 0.6445
Rm_Temperature:AWatering     1   5.8    0.0    0.0      B 0.8574
Rm_Temperature:ALine         9 100.8    1.4    1.4      B 0.2080
Rm_Temperature:AWatering:ALine  9 100.8    0.6    0.6      C 0.8095
Rm_Temperature:HWatering     1   6.4    0.2    0.2      B 0.7024
Rm_Temperature:HLine         9 114.3    2.8    2.8      B 0.0046
```

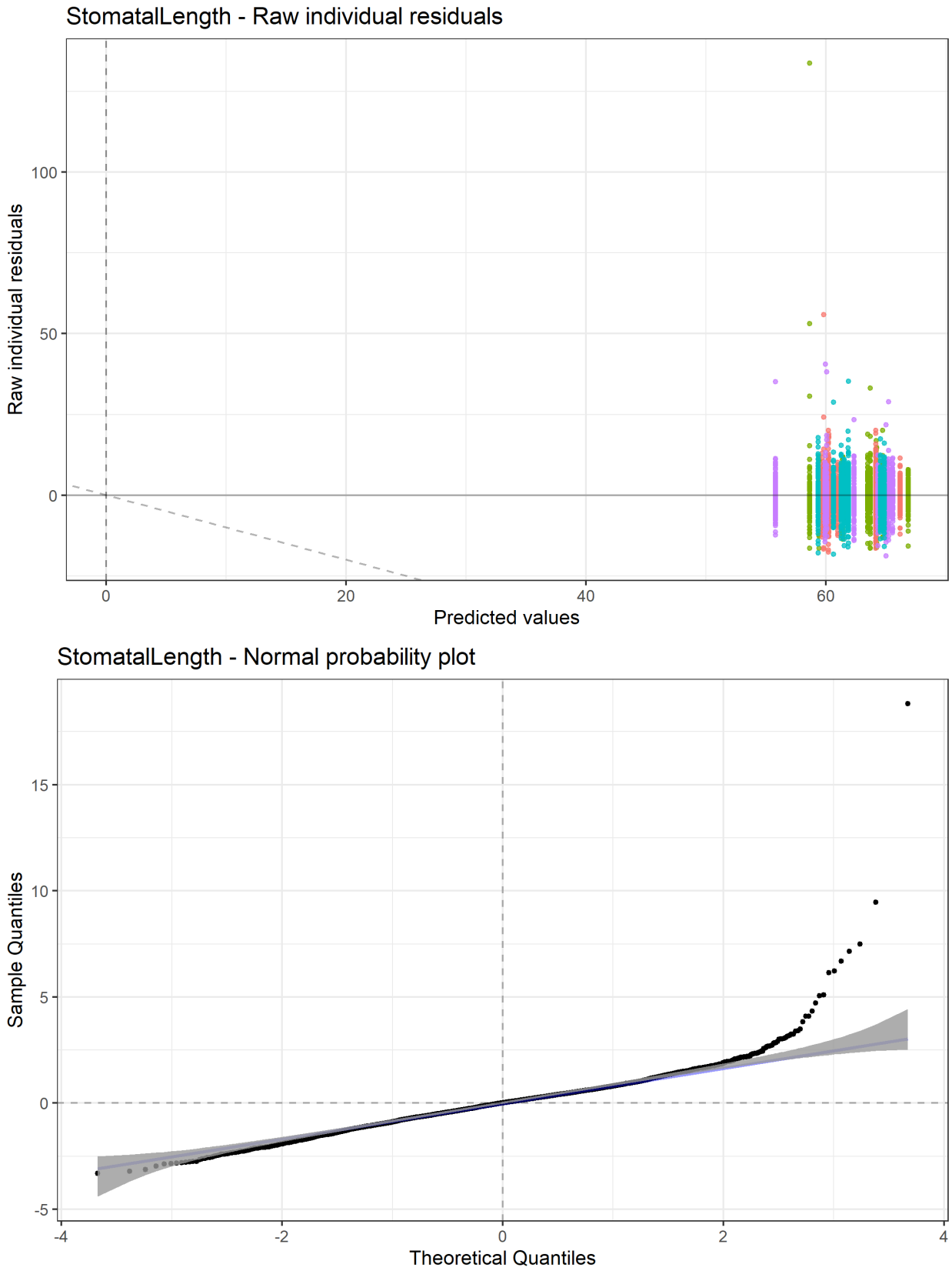


Figure 13: plot of chunk show-v1

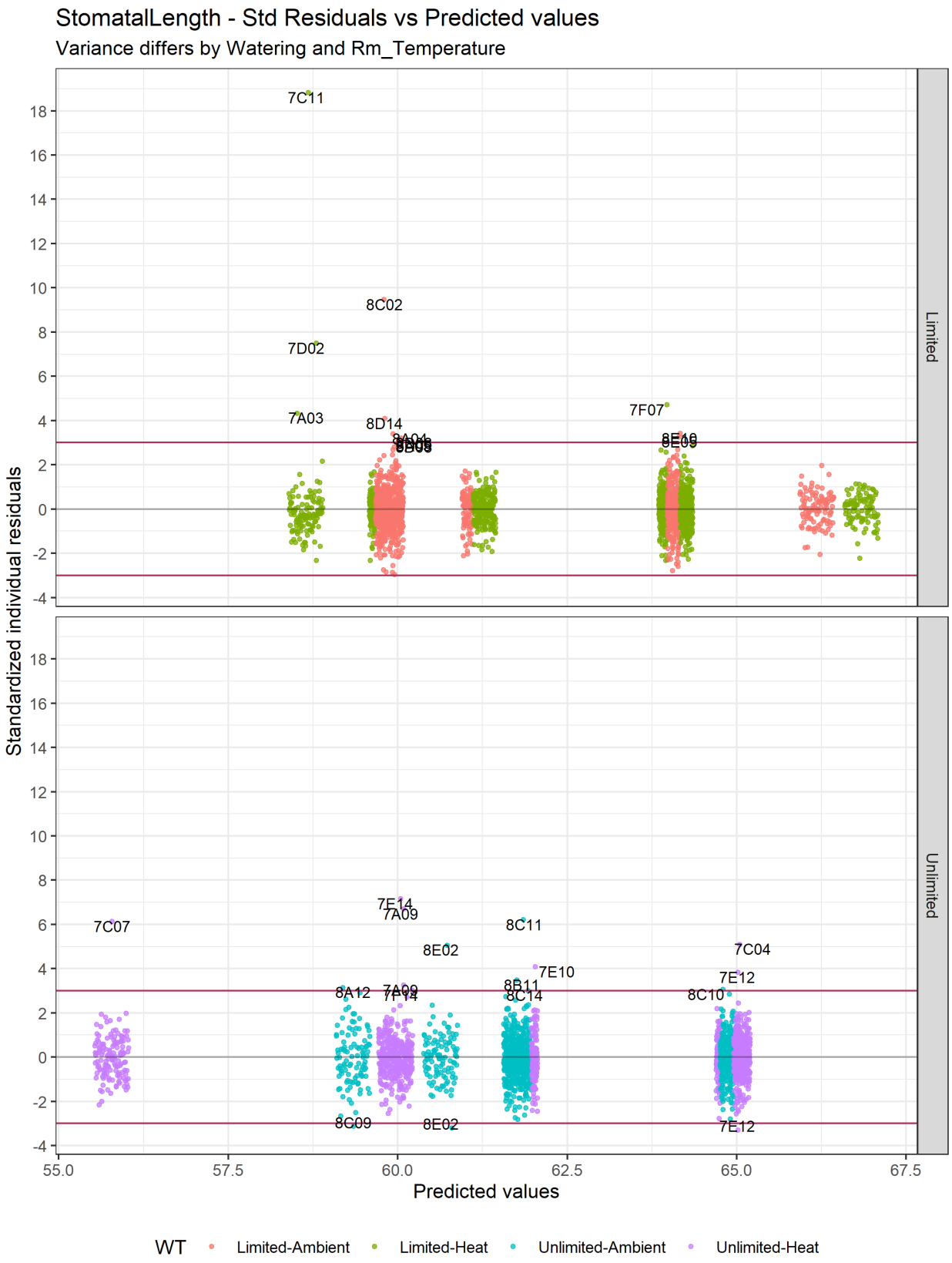


Figure 14: plot of chunk show-v1

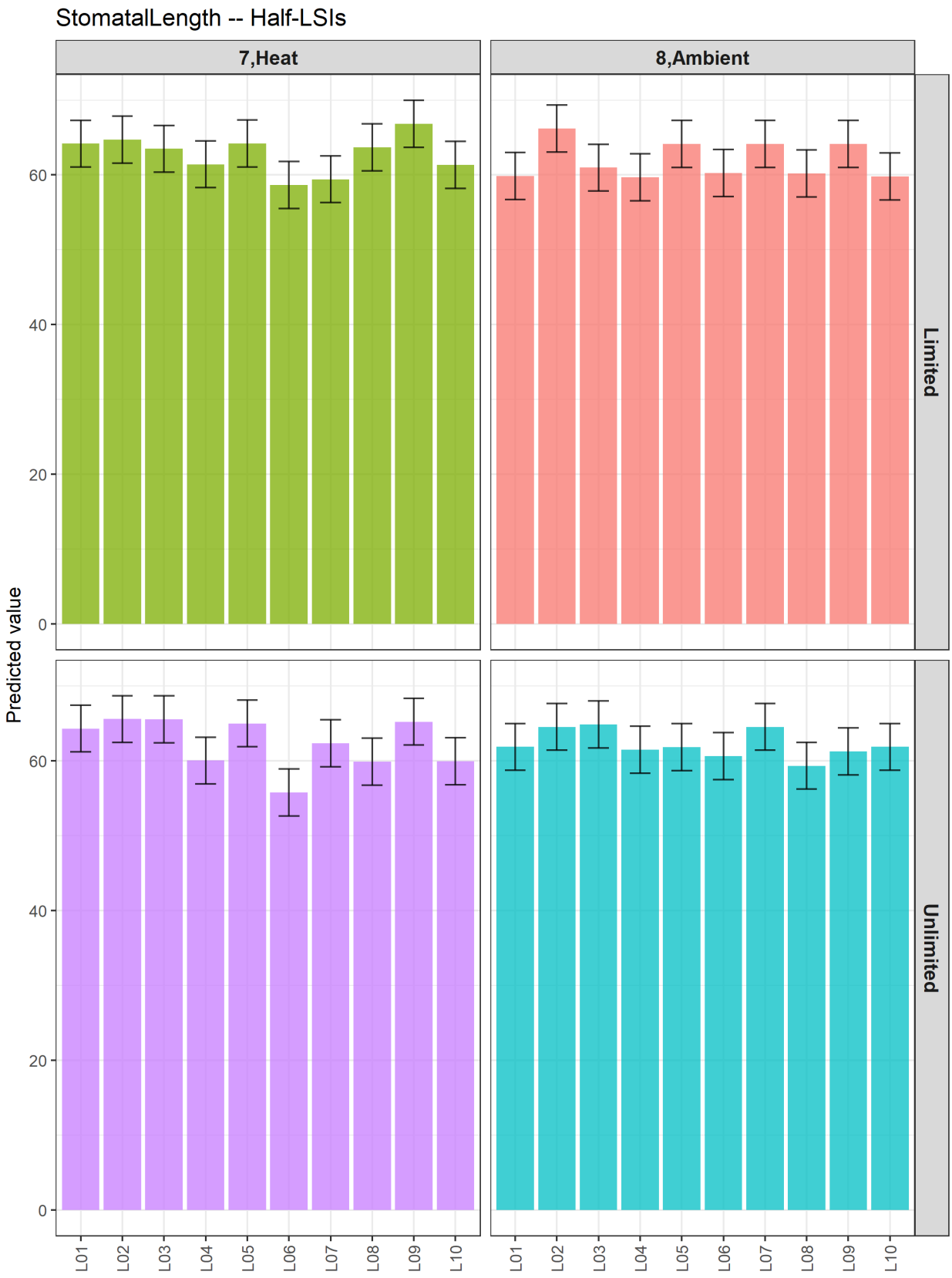


Figure 15: plot of chunk show-v1

```
Rm_Temperature:HWatering:HLine  9 117.2    0.6    0.6      C 0.7924


####  Sequence of model investigations for PoreArea

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

                                                                                terms DF denDF      p
1                                                                       Initial model 43   4.0     NA
2  Start with residual heterogeneity in Watering & Rm_Temperature for Residual NA    NA     NA
3                                Try homogeneity in Rm_Temperature for Residual  0  -2.0     NA
4                                      Try homogeneity in Watering for Residual  0  -2.0     NA
5                                                                   Room:Side  1    NA     NA
6                                                    Fix bound variance components  0   0.0     NA
7                                                  Rm_Temperature:HWatering:HLine  9 117.2 0.7924
8                                                            Rm_Temperature:HLine  9 114.3 0.0046
9                                                         Rm_Temperature:HWatering  1   6.4 0.7024
10                                                 Rm_Temperature:AWatering:ALine  9 100.8 0.8095
11                                                           Rm_Temperature:ALine  9 100.8 0.2080
12                                                        Rm_Temperature:AWatering  1   5.8 0.8574
          AIC        BIC           action
1  49530.2771 49828.2674   Starting model
2          NA         NA Changed residual
3    266.5488   253.8890        Unswapped
4    297.5733   284.9135        Unswapped
5          NA         NA         Boundary
6      0.0000     0.0000          Swapped
7          NA         NA   Nonsignificant
8          NA         NA      Significant
9          NA         NA   Nonsignificant
10         NA         NA   Nonsignificant
11         NA         NA   Nonsignificant
12         NA         NA   Nonsignificant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine"

LSDs
          c   minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 167.2631 177.5638 196.4971    177.5638    0.106628        2        2

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`


#### Results for gsmax


  Final model converged
```

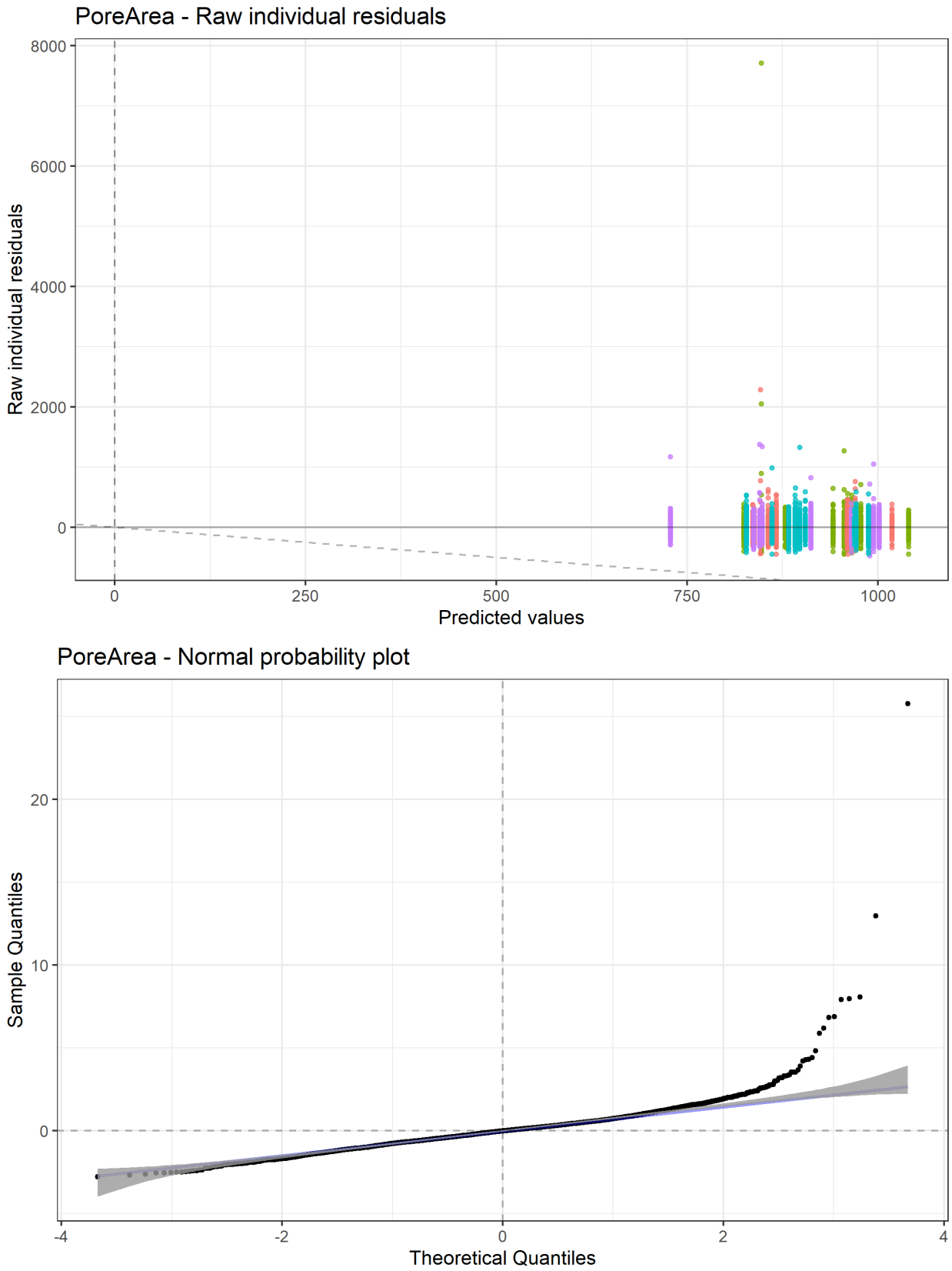


Figure 16: plot of chunk show-v1

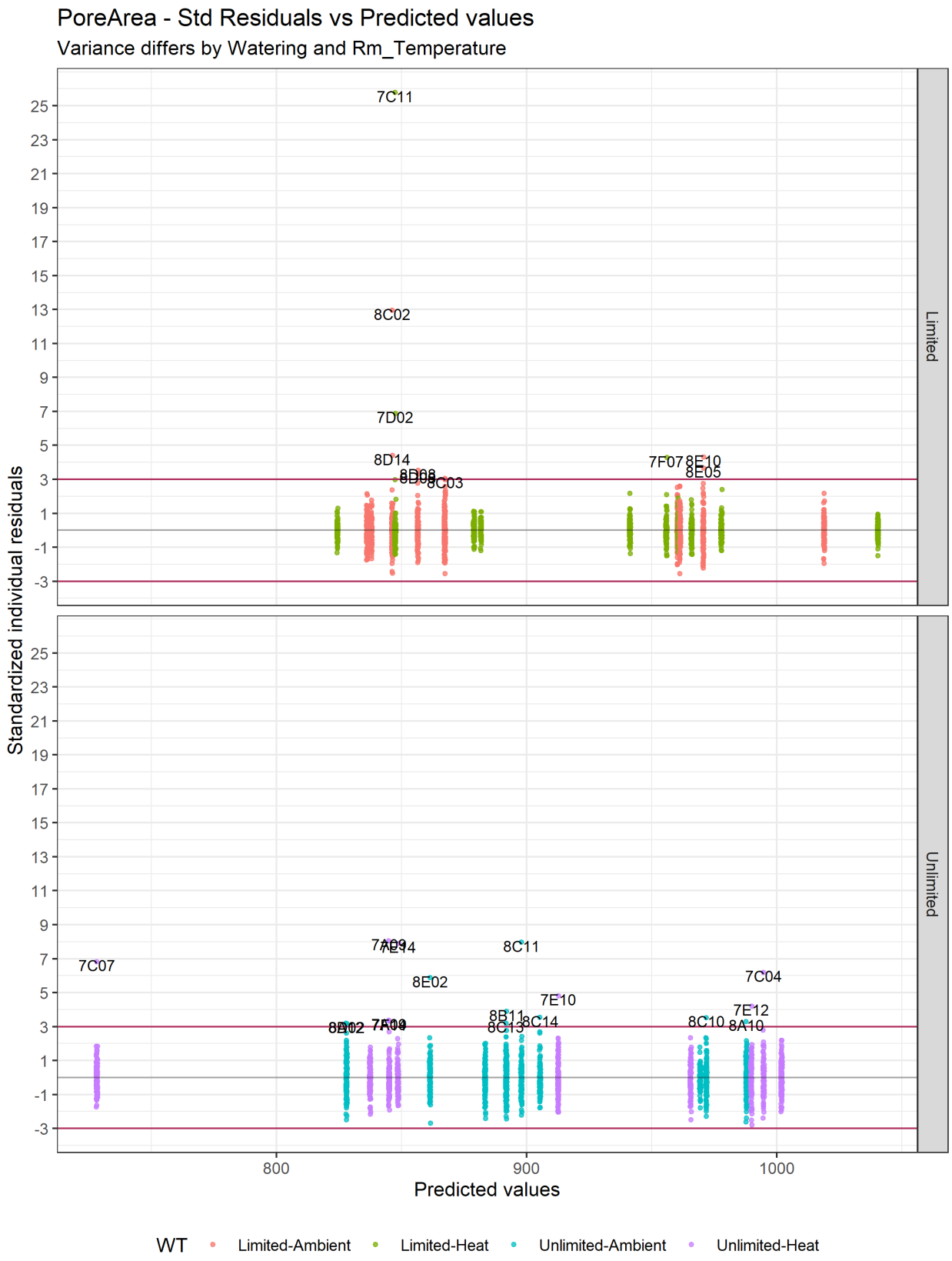


Figure 17: plot of chunk show-v1

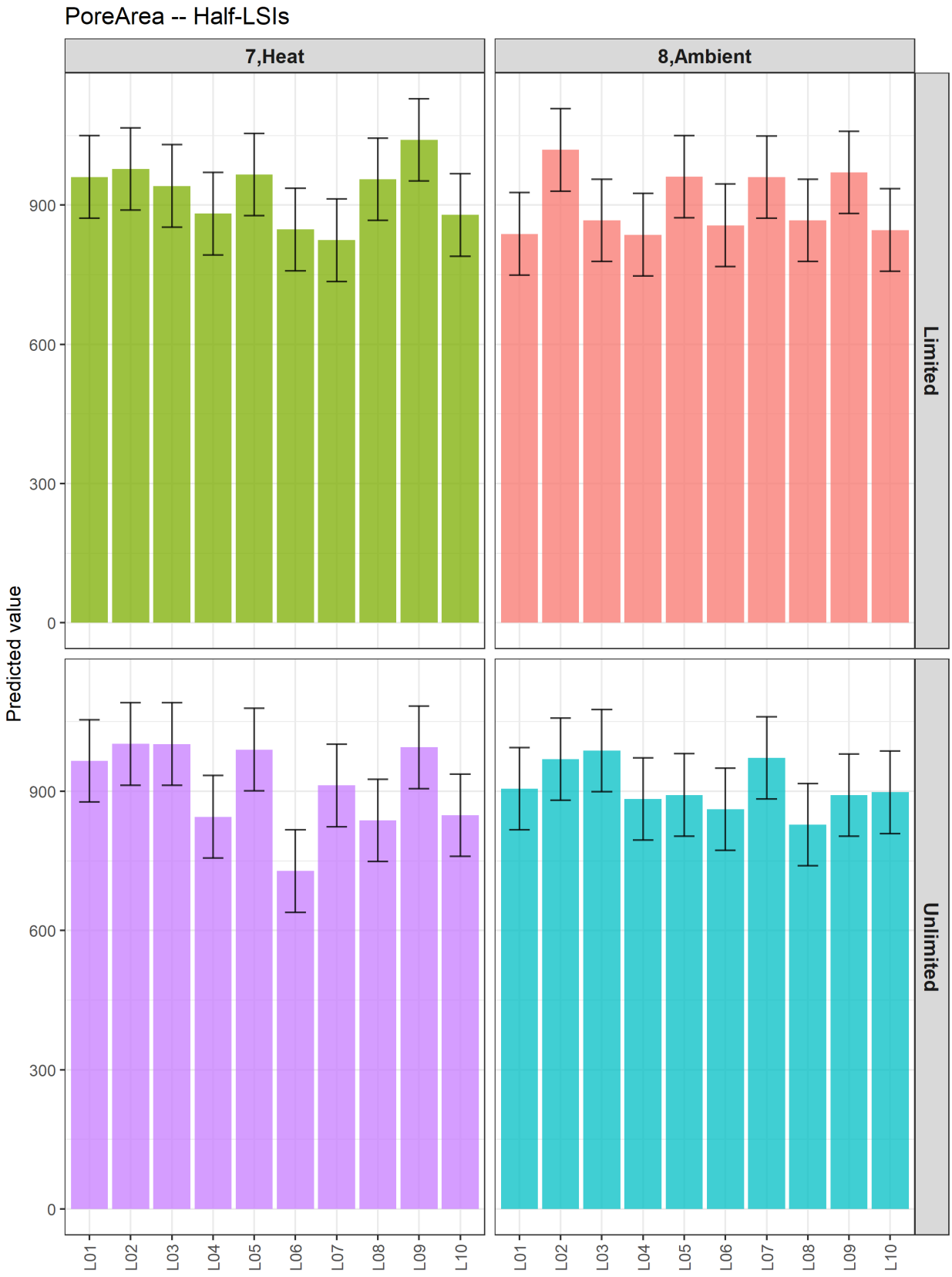


Figure 18: plot of chunk show-v1

```
#### Formulae from asreml object

fixed: gsmax ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine + HWatering *
     HLine)
random: ~ Room:Side:Location
residual: NULL




####  Summary of the fitted variance parameters for gsmax

                      component   std.error   z.ratio bound %ch
Room:Side:Location 0.0008122171 0.002065471 0.3932358     P   0
units!R            0.0336374469 0.004475327 7.5161981     P   0


####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: gsmax

                              Df denDF  F.inc  F.con Margin     Pr
(Intercept)                    1     3 5752.0 5752.0        0.0000
ASide                          1     3    3.1    3.0      B 0.1826
ASide:ABlock                   2     3    3.8    3.9      C 0.1468
Rm_Temperature                 1     3    0.4    0.4      A 0.5767
Rm_Temperature:AWatering       1   113    1.3    1.3      B 0.2581
Rm_Temperature:ALine           9   113    1.1    1.1      B 0.3636
Rm_Temperature:AWatering:ALine 9   113    1.8    1.8      C 0.0777
Rm_Temperature:HWatering       1   113    1.4    1.6      B 0.2153
Rm_Temperature:HLine           9   113    2.5    2.5      B 0.0106
Rm_Temperature:HWatering:HLine 9   113    0.7    0.7      C 0.7318


####  Sequence of model investigations for gsmax

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

                                                                    terms DF denDF      p
1                                                                Initial model 43     2     NA
2  Start with residual heterogeneity in Watering & Rm_Temperature for Residual NA    NA     NA
3                              Try homogeneity in Rm_Temperature for Residual  0    -2     NA
4                   Try homogeneity in Watering & Rm_Temperature for Residual  0    -1     NA
5                                                                 Room:Side  1    NA     NA
6                                                    Room:Side:Location:SubUnit  1    NA     NA
7                                                 Fix bound variance components  0     0     NA
8                                                Rm_Temperature:HWatering:HLine  9   113 0.7318
9                                                          Rm_Temperature:HLine  9   113 0.0106
10                                                       Rm_Temperature:HWatering  1   113 0.2153
11                                               Rm_Temperature:AWatering:ALine  9   113 0.0777
```

```
12                                                    Rm_Temperature:ALine  9   113 0.3636
13                                                Rm_Temperature:AWatering  1   113 0.2581
           AIC         BIC           action
1  -330.210392 -192.109703   Starting model
2           NA          NA Changed residual
3    -1.238629   -6.745809          Swapped
4    -1.724391   -4.477981          Swapped
5           NA          NA         Boundary
6           NA          NA         Boundary
7     0.000000    0.000000          Swapped
8           NA          NA   Nonsignificant
9           NA          NA      Significant
10          NA          NA   Nonsignificant
11          NA          NA   Nonsignificant
12          NA          NA   Nonsignificant
13          NA          NA   Nonsignificant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine"

LSDs
          c    minLSD   meanLSD  maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 0.2568615 0.2595447 0.28097   0.2595447  0.08254961        1        3

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`
```

MODEL SUMMARIES

Wald tables

```
for (resp in responses.anat[kvec]) {
  tbl <- models[[resp]]$wald.tab[-1,]
  cat("\n\n", resp)
  print(kable(tbl, digits = 3))
}
```

```
 StomatalCount

|                              | Df| denDF| F.inc| F.con|Margin |    Pr|
|:-----------------------------|--:|-----:|-----:|-----:|:------|-----:|
|ASide                         |  1| 101.6| 2.202| 2.017|B      | 0.159|
|ASide:ABlock                  |  2| 101.7| 5.658| 5.984|C      | 0.003|
|Rm_Temperature                |  1| 104.0| 0.079| 0.079|A      | 0.779|
|Rm_Temperature:AWatering      |  1|  58.9| 0.942| 0.942|B      | 0.336|
|Rm_Temperature:ALine          |  9|  58.9| 0.872| 0.872|B      | 0.555|
|Rm_Temperature:AWatering:ALine|  9|  58.9| 0.827| 0.827|C      | 0.594|
|Rm_Temperature:HWatering      |  1|  56.8| 0.910| 1.064|B      | 0.307|
|Rm_Temperature:HLine          |  9|  56.8| 7.074| 7.074|B      | 0.000|
```

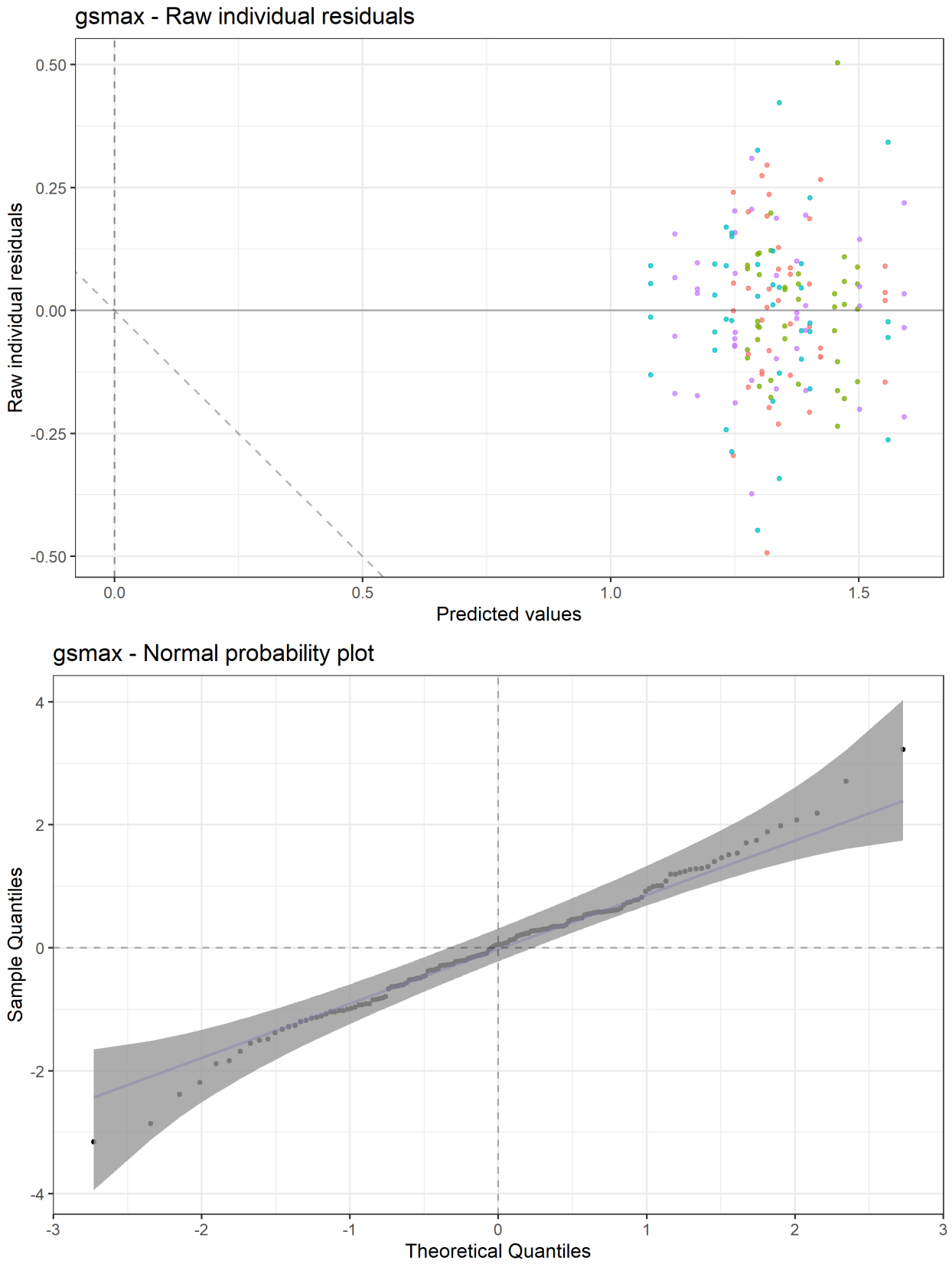


Figure 19: plot of chunk show-v1

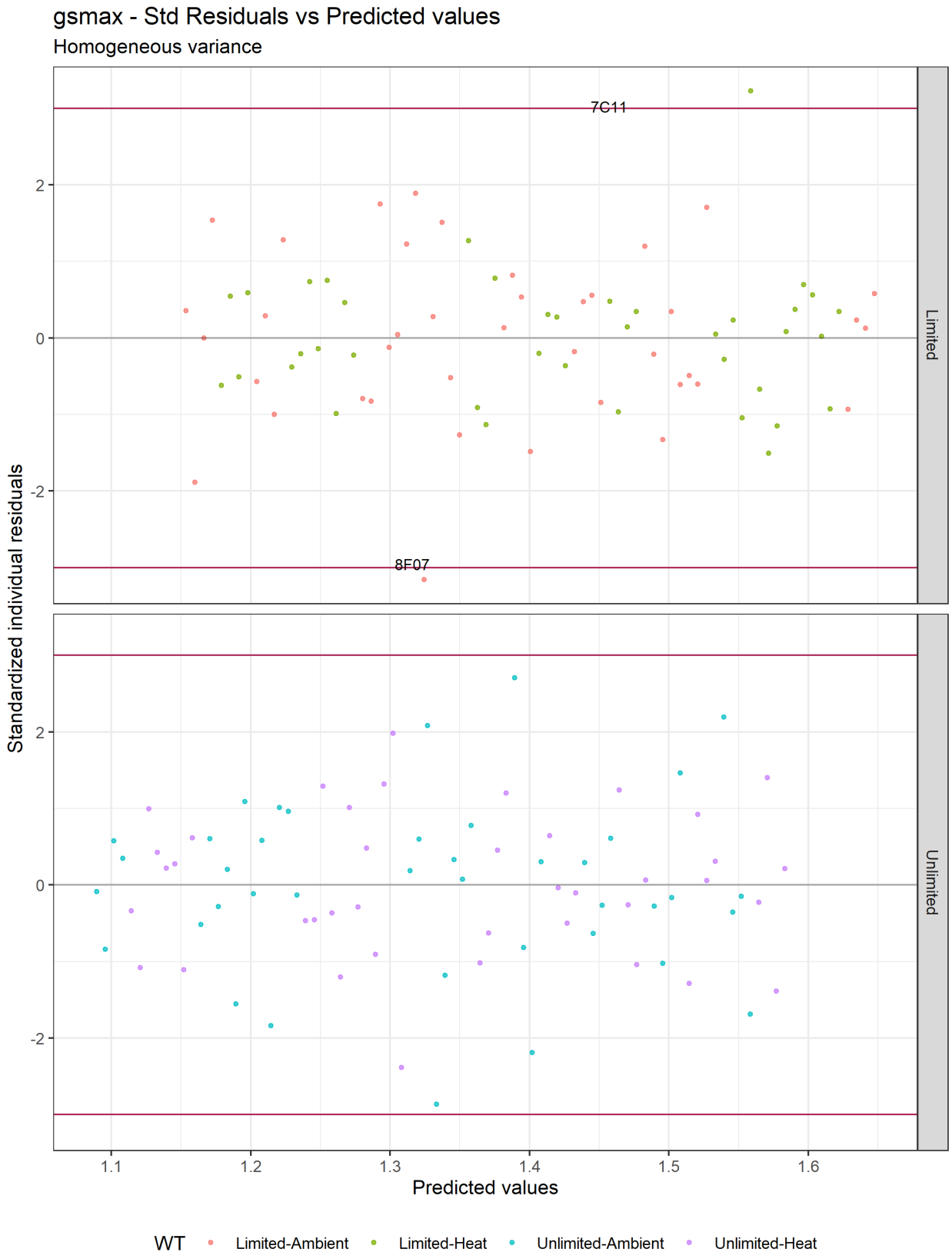


Figure 20: plot of chunk show-v1

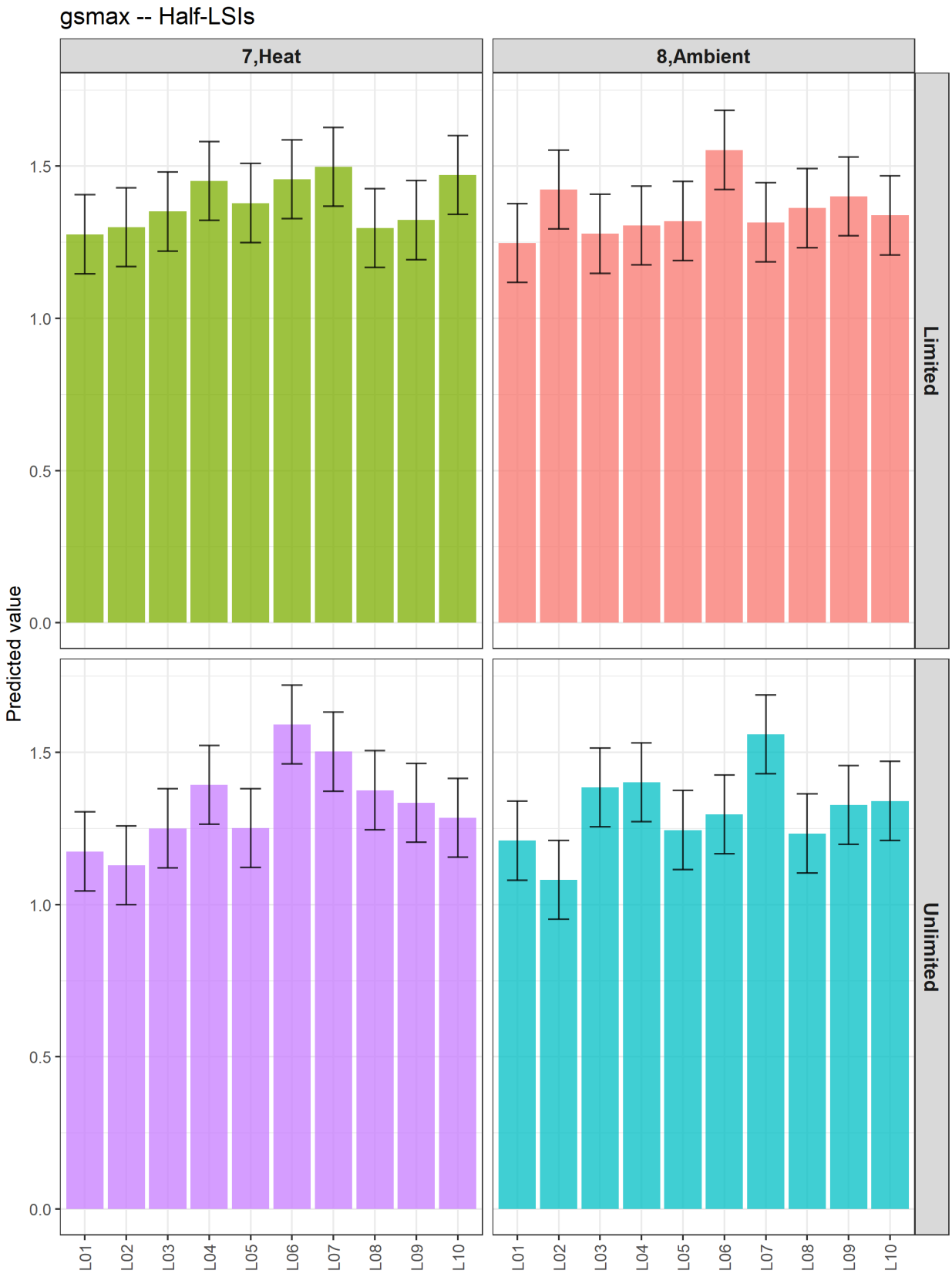


Figure 21: plot of chunk show-v1

|Rm_Temperature:HWatering:HLine | 9| 56.8| 1.692| 1.692|C | 0.112|

StomatalDensity

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	101.6	2.202	2.017	B	0.159
ASide:ABlock	2	101.7	5.658	5.984	C	0.003
Rm_Temperature	1	104.0	0.079	0.079	A	0.779
Rm_Temperature:AWatering	1	58.9	0.942	0.942	B	0.336
Rm_Temperature:ALine	9	58.9	0.872	0.872	B	0.555
Rm_Temperature:AWatering:ALine	9	58.9	0.827	0.827	C	0.594
Rm_Temperature:HWatering	1	56.8	0.910	1.064	B	0.307
Rm_Temperature:HLine	9	56.8	7.074	7.074	B	0.000
Rm_Temperature:HWatering:HLine	9	56.8	1.692	1.692	C	0.112

StomatalArea

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	1.0	2.702	2.554	B	0.356
ASide:ABlock	2	8.0	0.052	0.054	C	0.948
Rm_Temperature	1	1.0	0.431	0.431	A	0.630
Rm_Temperature:AWatering	1	7.9	2.788	2.801	B	0.133
Rm_Temperature:ALine	9	104.9	3.438	3.438	B	0.001
Rm_Temperature:AWatering:ALine	9	104.9	1.551	1.551	C	0.140
Rm_Temperature:HWatering	1	8.1	1.671	1.519	B	0.252
Rm_Temperature:HLine	9	109.2	3.321	3.321	B	0.001
Rm_Temperature:HWatering:HLine	9	109.3	0.303	0.303	C	0.972

StomatalWidth

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	1.0	2.405	2.225	B	0.376
ASide:ABlock	2	8.0	0.065	0.092	C	0.913
Rm_Temperature	1	1.0	0.312	0.312	A	0.676
Rm_Temperature:AWatering	1	7.9	8.409	8.446	B	0.020
Rm_Temperature:ALine	9	106.1	2.405	2.404	B	0.016
Rm_Temperature:AWatering:ALine	9	106.1	2.308	2.308	C	0.021
Rm_Temperature:HWatering	1	8.1	7.487	6.668	B	0.032
Rm_Temperature:HLine	9	107.6	3.178	3.178	B	0.002
Rm_Temperature:HWatering:HLine	9	107.7	0.285	0.285	C	0.978

StomatalLength

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	3.0	0.315	0.319	B	0.611
ASide:ABlock	2	3.0	0.600	0.607	C	0.601

Rm_Temperature	1	3.0	0.277	0.277	A	0.635
Rm_Temperature:AWatering	1	5.9	0.052	0.051	B	0.828
Rm_Temperature:ALine	9	105.5	1.454	1.454	B	0.175
Rm_Temperature:AWatering:ALine	9	105.5	0.505	0.505	C	0.868
Rm_Temperature:HWatering	1	6.1	0.133	0.103	B	0.759
Rm_Temperature:HLine	9	108.6	3.176	3.176	B	0.002
Rm_Temperature:HWatering:HLine	9	108.8	0.475	0.475	C	0.889

PoreArea

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	3.0	0.305	0.308	B	0.618
ASide:ABlock	2	3.0	0.582	0.588	C	0.609
Rm_Temperature	1	3.0	0.261	0.261	A	0.644
Rm_Temperature:AWatering	1	5.8	0.036	0.035	B	0.857
Rm_Temperature:ALine	9	100.8	1.378	1.378	B	0.208
Rm_Temperature:AWatering:ALine	9	100.8	0.582	0.582	C	0.810
Rm_Temperature:HWatering	1	6.4	0.189	0.160	B	0.702
Rm_Temperature:HLine	9	114.3	2.849	2.849	B	0.005
Rm_Temperature:HWatering:HLine	9	117.2	0.603	0.603	C	0.792

gsmax

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	3	3.103	2.983	B	0.183
ASide:ABlock	2	3	3.770	3.891	C	0.147
Rm_Temperature	1	3	0.390	0.390	A	0.577
Rm_Temperature:AWatering	1	113	1.292	1.292	B	0.258
Rm_Temperature:ALine	9	113	1.107	1.107	B	0.364
Rm_Temperature:AWatering:ALine	9	113	1.790	1.790	C	0.078
Rm_Temperature:HWatering	1	113	1.450	1.553	B	0.215
Rm_Temperature:HLine	9	113	2.546	2.546	B	0.011
Rm_Temperature:HWatering:HLine	9	113	0.673	0.673	C	0.732

Chosen terms

```
tableChosenModels(models, responses.anat)
```

```
  Trait
1 StomatalCount
2 StomatalDensity
3 StomatalArea
4 StomatalWidth
5 StomatalLength
6 PoreArea
7 gsmax
  Chosen.model
1 Rm_Temperature:HLine
2 Rm_Temperature:HLine
3 Rm_Temperature:HLine + Rm_Temperature:ALine
```

```
4 Rm_Temperature:HLine + Rm_Temperature:HWatering + Rm_Temperature:AWatering:ALine
5 Rm_Temperature:HLine
6 Rm_Temperature:HLine
7 Rm_Temperature:HLine
```

Table the residual variances

```
tableVarModels(models, responses.anat)
```

```
  Trait           Variance.model
1 StomatalCount   Variance differs by Rm_Temperature
2 StomatalDensity Variance differs by Rm_Temperature
3 StomatalArea    Variance differs by Rm_Temperature
4 StomatalWidth   Variance differs by Watering and Rm_Temperature
5 StomatalLength  Variance differs by Watering and Rm_Temperature
6 PoreArea        Variance differs by Watering and Rm_Temperature
7 gsmax           Homogeneous variance
```

Extremal residuals

```
typelabls <- c("Anat")
resptlabs <- rep(typelabls,
                 times = length(responses.anat))
print(cbind(responses.anat, resptlabs), quote=F)
```

```
     responses.anat  resptlabs
[1,] StomatalCount   Anat
[2,] StomatalDensity Anat
[3,] StomatalArea    Anat
[4,] StomatalWidth   Anat
[5,] StomatalLength  Anat
[6,] PoreArea        Anat
[7,] gsmax           Anat
```

```
res.lim.label <- 5
Outliers.dat <- data.frame()
tbl <- data.frame()
for (resp in responses.anat[kvec])
{
  if(resp %in% responses.pot){
    tmp.dat <- models[[resp]]$asreml.obj$call$data
    res.dat <- within(tmp.dat, res.std <- residuals(models[[resp]]$asreml.obj, type="stdCond"))
    res.dat <- res.dat[!is.na(res.dat$res.std), ]
    outliers.dat <- subset(res.dat, abs(res.std) > res.lim.label)
    outliers.dat <- outliers.dat[c("Cell.ID", "res.std", "Rm_Temperature",
                                   "Watering", "Line")]
    if (nrow(outliers.dat) > 0)
    {
      Outliers.dat <- rbind(Outliers.dat,
                            data.frame(Response=resp, outliers.dat))
      tbl <- rbind(tbl,
                   data.frame(Response = resp,
                              n        = nrow(outliers.dat),
                              rmin     = min(res.dat$res.std, na.rm=T),
```

```
                              rmax     = max(res.dat$res.std, na.rm=T)))
    }
  } else{
    tmp.dat <- models[[resp]]$asreml.obj$call$data
    res.dat <- within(tmp.dat, res.std <- residuals(models[[resp]]$asreml.obj, type="stdCond"))
    res.dat <- res.dat[!is.na(res.dat$res.std), ]
    outliers.dat <- subset(res.dat, abs(res.std) > res.lim.label)
    outliers.dat <- outliers.dat[c("Cell.ID", "Stomata", "res.std", "Rm_Temperature",
                                   "Watering", "Line")]
    if (nrow(outliers.dat) > 0)
    {
      Outliers.dat <- rbind(Outliers.dat,
                            data.frame(Response=resp, outliers.dat))
      tbl <- rbind(tbl,
                   data.frame(Response = resp,
                              n        = nrow(outliers.dat),
                              rmin     = min(res.dat$res.std, na.rm=T),
                              rmax     = max(res.dat$res.std, na.rm=T)))
    }
  }
}

if (nrow(Outliers.dat)==0) {

  cat("\n\n #### No outliers found\n\n")

} else {

  Outliers.dat <- with(Outliers.dat, Outliers.dat[order(Cell.ID, Stomata, Response),])
  Outliers.dat <- within(Outliers.dat, res <- round(res.std,2))
  tmp.dat <- Outliers.dat[c("Cell.ID","Stomata", "res","Response", "Rm_Temperature",
                            "Watering", "Line")]
  names(tmp.dat)[1] <- c("Plant")
  outliers.dat <- unique(tmp.dat[,-(3:4)])
  tbl <- within(tbl, {
    rmin <- ifelse(rmin < 0, round(rmin,2), NA)
    rmax <- ifelse(rmax > 0, round(rmax,2), NA)
  })

  msg1 <- paste("Standardised residuals outside the range (-",
                res.lim.label, ", ",res.lim.label,")",
                sep="")
  msg2 <- "Unique plants in outlier list"

  cat("\n\n #### Number of outliers per response\n\n"); print(tbl, right=F)
  cat("\n\n",msg1,"\n\n"); print(tmp.dat, right=F)
  cat("\n\n",msg2,"\n\n"); print(outliers.dat, right=F)

  tmp2.dat <- within(tmp.dat, {
    Magnitude <- abs(res)
    Direction <- factor(sign(res), levels = c(1,-1), labels = c("pos","neg"))
    Response <- factor(Response, levels = responses.anat)
    Resptype <- dae::fac.recode(Response, newlevels = resptlabs, levels = rev(typelabls))
```

```
  })

  lims <- c(res.lim.label,
            max(4, 0.5*ceiling(max(tmp2.dat$Magnitude)/0.5)))  #round to nearest half
  plt <- ggplot(tmp2.dat, aes(Plant, as.character(Response),
                              shape=Direction, colour=Magnitude, fill=Magnitude)) +
    facet_grid(rows = vars(Resptype), space="free", scales="free") +
    geom_point(size=2) +
    scale_colour_gradient(low="green", high="red", limits = lims) +
    scale_fill_gradient(low="green", high="red", limits = lims) +
    scale_shape_manual(values = c(24,25), drop=F) +
    labs(y=NULL) +
    scale_y_discrete(drop=F) +
    ggtitle("Outliers by Plant and Response",
            subtitle = "Coloured by magnitude of standardized residual") +
    theme_bw() +
    theme(axis.text.x = element_text(angle=90, hjust=1, vjust=0.5),
          strip.text = element_text(size=10, face="bold"))

  print(plt)
}
```

```
 #### Number of outliers per response

  Response       n  rmin  rmax
1 StomatalArea   13 -4.55 16.06
2 StomatalWidth   5 -4.81 10.55
3 StomatalLength  9 -3.30 18.82
4 PoreArea        9 -2.80 25.78


 Standardised residuals outside the range (-5, 5)

      Plant   Stomata res   Response       Rm_Temperature Watering
19273 CER7A09 30       8.05 PoreArea       7,Heat         Unlimited
1927  CER7A09 30       8.16 StomatalArea   7,Heat         Unlimited
19272 CER7A09 30       6.69 StomatalLength 7,Heat         Unlimited
19271 CER7A09 30       5.06 StomatalWidth  7,Heat         Unlimited
4108  CER7A12 19       6.32 StomatalWidth  7,Heat         Unlimited
41322 CER7C04 24       6.18 PoreArea       7,Heat         Unlimited
4132  CER7C04 24       6.57 StomatalArea   7,Heat         Unlimited
41321 CER7C04 24       5.10 StomatalLength 7,Heat         Unlimited
28262 CER7C07 39       6.82 PoreArea       7,Heat         Unlimited
2826  CER7C07 39       5.13 StomatalArea   7,Heat         Unlimited
28261 CER7C07 39       6.14 StomatalLength 7,Heat         Unlimited
27051 CER7C11 38      25.78 PoreArea       7,Heat         Limited
2705  CER7C11 38      18.82 StomatalLength 7,Heat         Limited
27343 CER7D02 29       6.88 PoreArea       7,Heat         Limited
2734  CER7D02 29      16.06 StomatalArea   7,Heat         Limited
27342 CER7D02 29       7.49 StomatalLength 7,Heat         Limited
27341 CER7D02 29      10.55 StomatalWidth  7,Heat         Limited
4342  CER7E14 21       7.91 PoreArea       7,Heat         Unlimited
```

434	CER7E14	21	6.08	StomatalArea	7,Heat	Unlimited	
4341	CER7E14	21	7.16	StomatalLength	7,Heat	Unlimited	
3884	CER8A03	32	5.55	StomatalArea	8,Ambient	Limited	
38841	CER8A03	32	6.67	StomatalWidth	8,Ambient	Limited	
792	CER8C02	30	12.95	PoreArea	8,Ambient	Limited	
79	CER8C02	30	5.22	StomatalArea	8,Ambient	Limited	
791	CER8C02	30	9.47	StomatalLength	8,Ambient	Limited	
2969	CER8C05	28	5.43	StomatalArea	8,Ambient	Limited	
1762	CER8C11	20	7.96	PoreArea	8,Ambient	Unlimited	
176	CER8C11	20	6.20	StomatalArea	8,Ambient	Unlimited	
1761	CER8C11	20	6.22	StomatalLength	8,Ambient	Unlimited	
25482	CER8E02	15	5.87	PoreArea	8,Ambient	Unlimited	
2548	CER8E02	15	5.08	StomatalArea	8,Ambient	Unlimited	
25481	CER8E02	15	5.06	StomatalLength	8,Ambient	Unlimited	
3819	CER8E05	19	5.65	StomatalArea	8,Ambient	Limited	
1340	CER8E12	27	5.42	StomatalArea	8,Ambient	Unlimited	
13401	CER8E12	27	6.82	StomatalWidth	8,Ambient	Unlimited	
156	CER8F10	28	5.10	StomatalArea	8,Ambient	Unlimited	

	Line
19273	PBI17N013-0C-0N-0N-010N-8N
1927	PBI17N013-0C-0N-0N-010N-8N
19272	PBI17N013-0C-0N-0N-010N-8N
19271	PBI17N013-0C-0N-0N-010N-8N
4108	PBI17N016-0C-0N-0N-010N-27N
41322	PBI17N016-0C-0N-0N-010N-27N
4132	PBI17N016-0C-0N-0N-010N-27N
41321	PBI17N016-0C-0N-0N-010N-27N
28262	PBI16C009-0C-0N-0N-010N-14N
2826	PBI16C009-0C-0N-0N-010N-14N
28261	PBI16C009-0C-0N-0N-010N-14N
27051	PBI16C009-0C-0N-0N-010N-14N
2705	PBI16C009-0C-0N-0N-010N-14N
27343	PBI16C009-0C-0N-0N-010N-14N
2734	PBI16C009-0C-0N-0N-010N-14N
27342	PBI16C009-0C-0N-0N-010N-14N
27341	PBI16C009-0C-0N-0N-010N-14N
4342	PBI17N002-0C-0N-0N-010N-16N
434	PBI17N002-0C-0N-0N-010N-16N
4341	PBI17N002-0C-0N-0N-010N-16N
3884	PBI17N016-0C-0N-0N-010N-27N
38841	PBI17N016-0C-0N-0N-010N-27N
792	PBI17N002-0C-0N-0N-010N-16N
79	PBI17N002-0C-0N-0N-010N-16N
791	PBI17N002-0C-0N-0N-010N-16N
2969	PBI19N007-0N-3N
1762	PBI17N002-0C-0N-0N-010N-16N
176	PBI17N002-0C-0N-0N-010N-16N
1761	PBI17N002-0C-0N-0N-010N-16N
25482	PBI16C009-0C-0N-0N-010N-14N
2548	PBI16C009-0C-0N-0N-010N-14N
25481	PBI16C009-0C-0N-0N-010N-14N
3819	PBI17N016-0C-0N-0N-010N-27N
1340	PBI17N006-0N-0N-89N
13401	PBI17N006-0N-0N-89N

156 PBI17N002-0C-0N-0N-010N-16N

Unique plants in outlier list

	Plant	Stomata	Rm_Temperature	Watering	Line
19273	CER7A09	30	7,Heat	Unlimited	PBI17N013-0C-0N-0N-010N-8N
4108	CER7A12	19	7,Heat	Unlimited	PBI17N016-0C-0N-0N-010N-27N
41322	CER7C04	24	7,Heat	Unlimited	PBI17N016-0C-0N-0N-010N-27N
28262	CER7C07	39	7,Heat	Unlimited	PBI16C009-0C-0N-0N-010N-14N
27051	CER7C11	38	7,Heat	Limited	PBI16C009-0C-0N-0N-010N-14N
27343	CER7D02	29	7,Heat	Limited	PBI16C009-0C-0N-0N-010N-14N
4342	CER7E14	21	7,Heat	Unlimited	PBI17N002-0C-0N-0N-010N-16N
3884	CER8A03	32	8,Ambient	Limited	PBI17N016-0C-0N-0N-010N-27N
792	CER8C02	30	8,Ambient	Limited	PBI17N002-0C-0N-0N-010N-16N
2969	CER8C05	28	8,Ambient	Limited	PBI19N007-0N-3N
1762	CER8C11	20	8,Ambient	Unlimited	PBI17N002-0C-0N-0N-010N-16N
25482	CER8E02	15	8,Ambient	Unlimited	PBI16C009-0C-0N-0N-010N-14N
3819	CER8E05	19	8,Ambient	Limited	PBI17N016-0C-0N-0N-010N-27N
1340	CER8E12	27	8,Ambient	Unlimited	PBI17N006-0N-0N-89N
156	CER8F10	28	8,Ambient	Unlimited	PBI17N002-0C-0N-0N-010N-16N

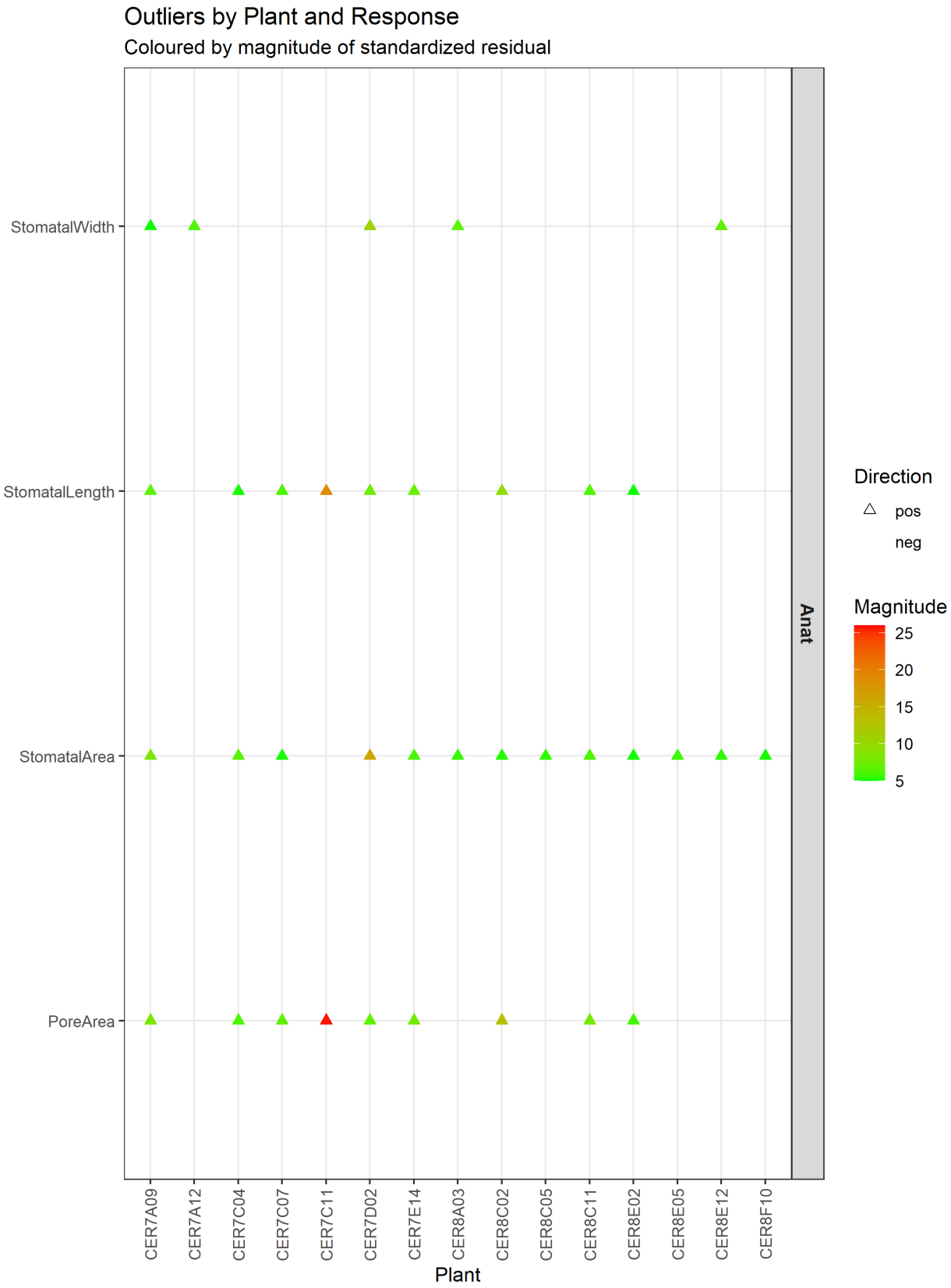


Figure 22: plot of chunk outliers-v1